Global regularity for 3D Navier–Stokes via a Gate / No-Theft critical-element rigidity spine

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Abstract

We prove global regularity for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3$ with smooth divergence-free initial data. The proof is organized as a critical-element rigidity argument at the scale-invariant L^3 level. Assuming a first finite-time singularity, we extract (via the concentration–compactness critical-element program [GKP13, Thm. 5–7]) an ancient suitable solution in Leray similarity variables whose orbit is precompact modulo translations. We then introduce a Gaussian "width" observable and establish a quantitative large-radius dichotomy for defect transport: either an absorption regime holds at a fixed scale, or a forced **Gate** mechanism produces a uniform dissipation packet. A localized Ornstein–Uhlenbeck adjoint Gaussian budget, combined with a No-Theft principle (pressure-tail decoupling), converts persistent Gate activation into finite total renormalized dissipation and hence asymptotic stationarity of the renormalization orbit. The resulting stationary Leray profile is excluded by Tsai's rigidity theorem for suitable backward self-similar solutions, yielding the contradiction.

Keywords: Navier–Stokes; regularity; critical element; concentration compactness; local energy inequality; rigidity.

MSC 2020: 35Q30, 76D03.

0. Main theorem and proof structure

0.1 Main theorem and setting

We consider the incompressible Navier–Stokes system on $\mathbb{R}^3$

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0,$$

with smooth divergence-free initial data $u_0 \in C_c^\infty(\mathbb{R}^3)$. (Viscosity is normalized to 1.)

Theorem 1.1 (Global regularity). For every such u_0, the corresponding Leray–Hopf solution is smooth for all $t > 0$. Equivalently, no finite-time singularity can occur.

0.2 Proof roadmap (one line per step)

Assume a first blow-up time T^*. Then:

1. **(CE extraction)** Use CE(L^3) to obtain an ancient suitable LRNS orbit (U, P) with critical bound and tightness (§2–§1).
2. **(Flux Sealing)** Prove a large-radius cutoff-flux barrier using critical compactness + Calderón–Zygmund (NFS.EST.4) pressure control (§2).
3. **(Modulation cost)** Build a Gaussian width observable and show pinned gauge forces non-degeneracy (§3).

4. **(Gaussian budget)** Derive the OU*-adjoint Gaussian energy inequality in similarity variables and pass to $R \to \infty$ (§4).
5. **(Gate / No-Theft dichotomy)** Show that persistent non-absorption triggers Gate packets; No-Theft turns packets into a finite dissipation budget (§4.1–§4.4).
6. **(Rigidity)** Finite renormalized dissipation forces asymptotic stationarity; the stationary profile contradicts Tsai ([Tsai98, Thm. 2]), and nontriviality is isolated in a single canonical lemma (Lemma 7.6P) (§4.5–§4.6).

This completes the contradiction and proves the main theorem.

0.3 Dependency and citation map

The manuscript is written as a single proof spine. The external inputs are limited and are recorded in statement form (with precise theorem identifiers) in Appendix **NFS.IMP**. Concretely:

- **Critical-element extraction at the L^3 level** (CE(L^3), Theorem 2.1), based on [GKP13, Thm. 5–7].
  From a hypothetical first blow-up time one extracts an ancient suitable orbit in Leray similarity variables with $L^\infty_\tau L^3_y$ control and tightness modulo translations.
- **Rigidity for backward self-similar suitable solutions** (Tsai) [Tsai98, Thm. 2].
  A stationary suitable backward self-similar profile must be trivial under Tsai's local energy finiteness hypothesis.

All remaining steps are proved in-line from these inputs using the local energy inequality, pressure localization, and the Gate/No-Theft mechanism. Some calculus identities in the change-of-variables and normalization steps admit straightforward symbolic verification (not included in this public RC).

For convenience, we list below the external inputs and where they are invoked, and we indicate which items are load-bearing versus toolbox references.

This canonical spine uses a **small set of external (citable) results**. Each is used only after we verify that its **hypotheses match** the solution class produced in §2–§4. Legend: † = load-bearing external input; ‡ = toolbox (Appendix NFS.EST) reference; **(ctx)** = context only (see References).

External inputs used (and where). 1. **Suitable weak solutions / LEI formalism.** Definition + compactness framework: [CKN], [LR02]. *Used in:* §1.1, §2–§4, and whenever we pass to limits. 2. **Pressure representation and Calderón–Zygmund (NFS.EST.4) bounds.** $p = \mathcal{R}_i\mathcal{R}_j(u_iu_j)$ and $L^{3/2}$ control from $u \in L^3$: [Gra14, Cor. 5.2.8] (hence $\mathcal{R}_i\mathcal{R}_j$ is bounded on $L^{3/2}$ by composition). *Used in:* §2 (Flux Sealing) and in Gaussian budgets (§4). 3. **Type I (backward self-similar) nonexistence under LEI.** [Tsai98] (load-bearing). Historical precursor: [NRS96] (context only; stronger assumptions; not used below). *Used in:* §6 (rigidity endpoint). 4. **Critical element extraction in L^3.** Minimal blow-up object / precompactness up to symmetries: [GKP13], [JS13]. *Used in:* §2 (to define the package (GKP1)–(GKP4)). We invoke the endpoint critical-element program at the L^3 level as developed by Gallagher–Koch–Planchon [GKP13, Thm. 5–7]. For the reader's convenience, we isolate exactly the extracted consequences we need in a single named package **CE(L^3)** (Theorem 2.1), with itemized hypotheses and outputs. A theorem-by-theorem hypothesis/usage map is recorded in Appendix **NFS.IMP** (Imported results) and Appendix **NFS.CE**.

This is the only place in the proof where concentration–compactness / minimal-element machinery

enters; all subsequent steps are proved in-line from the CE(L^3) package using the local energy inequality, pressure localization, and the Gate/No-Theft mechanism.

Theorem 2.1 (CE(L^3) extraction in LRNS variables; contradiction output). Assume for contradiction that there exists a Leray–Hopf solution of 3D Navier–Stokes with a first singular time $T^* < \infty$. Then one can extract (after blow-up rescaling and canonical symmetry pinning) an **ancient LRNS suitable solution** (U, P) on $(-\infty, 0] \times \mathbb{R}^3$ such that:

(CE1) Critical bound. $U \in L^\infty_\tau L^3_y(\mathbb{R}^3)$ with $\sup_{\tau \le 0} \|U(\tau)\|_{L^3} \le M < \infty$.

(CE2) Uniform L^3-tightness (pinned). For every $\varepsilon > 0$ there exists $R_\varepsilon \ge 1$ such that

$$\sup_{\tau \le 0} \int_{|y| > R_\varepsilon} |U(\tau, y)|^3 \, dy \;\le\; \varepsilon. \tag{CE2}$$

(CE3) Suitability in LRNS (LRNS-LEI). (U, P) is a suitable weak solution of (LRNS) on $(-\infty, 0] \times \mathbb{R}^3$ and satisfies the LRNS local energy inequality for every nonnegative test function $\varphi \in C^\infty_c(\mathbb{R}^3 \times (-\infty, 0])$.

(CE4) Pressure representation and localization. P may be chosen (distributionally) as $P = \mathcal{R}_i \mathcal{R}_j (U_i U_j)$, and the local pressure decomposition (NFS.EST.5) (NFS.EST.5) on balls holds on each bounded cylinder.

(CE5) Canonical pinning. Translations are fixed so that the tightness radius in (CE2) is measured relative to the origin.

Citation-level proof spine. The package above follows from the endpoint critical-element program: [GKP13, Thm. 5–7] provide existence/compactness/rigidity of critical elements, yielding the endpoint criterion $u \in L^\infty_t L^3_x \Rightarrow$ no finite-time blow-up (in the suitable framework). Suitability/LEI stability and pressure localization are in the CKN/ESS framework; see [CKN], [ESS03]. A theorem-number hypothesis map CE1–CE5 is recorded in Appendix **NFS.CE**.

Lemma 2.2 (Precompactness implies uniform L^3-tightness). Let $K \subset L^3(\mathbb{R}^3)$ be a **compact** set. Then for every $\varepsilon > 0$ there exists $R_\varepsilon \ge 1$ such that

$$\sup_{f \in K} \int_{|y| > R_\varepsilon} |f(y)|^3 \, dy \;\le\; \varepsilon. \tag{2.2}$$

In particular, if a trajectory $\{U(\tau)\}_{\tau \le 0} \subset L^3$ is precompact (after fixing translations), then (CE2) follows for the pinned orbit.

Proof. For $R \ge 1$ define $T_R(f) := \int_{|y|>R} |f(y)|^3 \, dy$. For fixed R, T_R is continuous on L^3. For fixed $f \in L^3$, $T_R(f) \to 0$ as $R \to \infty$ by absolute continuity of the integral. Fix $\varepsilon > 0$. For each $f \in K$ choose $R(f)$ so that $T_{R(f)}(f) < \varepsilon/2$. By continuity, there exists an L^3-neighborhood N_f such that $T_{R(f)}(g) < \varepsilon$ for all $g \in N_f$. A finite subcover $K \subset \bigcup_{j=1}^J N_{f_j}$ exists by compactness. Let $R_\varepsilon := \max_j R(f_j)$. Then for any $g \in K$, $T_{R_\varepsilon}(g) \le T_{R(f_j)}(g) < \varepsilon$ for some j, proving (2.2). $\square$

Lemma 2.3 (Suitability stability). Let $(u^{(n)}, p^{(n)})$ be suitable weak solutions of Navier–Stokes on a cylinder Q. Assume (after passing to a subsequence) that

$$u^{(n)} \to u \text{ in } L^3_{\mathrm{loc}}(Q), \qquad u^{(n)} \rightharpoonup u \text{ in } L^2_{\mathrm{loc}}(t; H^1_{\mathrm{loc}}(x)), \qquad p^{(n)} \rightharpoonup p \text{ in } L^{3/2}_{\mathrm{loc}}(Q).$$

Then (u, p) is a suitable weak solution on Q, and the local energy inequality passes to the limit.

Proof. Fix a nonnegative test function $\varphi \in C_c^\infty(Q)$. Each term in the local energy inequality passes to the limit under the above convergences; the dissipation term uses weak lower semicontinuity and $\varphi \geq 0$, while the remaining terms use the strong L^3_{loc} convergence. A term-by-term matching is recorded in Lemma 2.4. □

Lemma 2.4 (LEI term-by-term stability under L^3 strong convergence). Under the hypotheses of Lemma 2.3, for every nonnegative $\varphi \in C_c^\infty(Q)$ we have

$$\liminf_{n\to\infty} \int_Q \varphi\, |\nabla u^{(n)}|^2 \;\geq\; \int_Q \varphi\, |\nabla u|^2, \tag{2.4a}$$

and

$$\int_Q \tfrac{1}{2}|u^{(n)}|^2(\partial_t\varphi+\Delta\varphi)+\int_Q \left(\tfrac{1}{2}|u^{(n)}|^2+p^{(n)}\right) u^{(n)}\cdot\nabla\varphi \;\longrightarrow\; \int_Q \tfrac{1}{2}|u|^2(\partial_t\varphi+\Delta\varphi)+\int_Q \left(\tfrac{1}{2}|u|^2+p\right) u\cdot\nabla\varphi. \tag{2.4b}$$

Consequently the local energy inequality passes to the limit (Lemma 2.4).

Proof. Since $u^{(n)} \to u$ in $L^3_{\mathrm{loc}}(Q)$, we have $|u^{(n)}|^2 \to |u|^2$ in $L^{3/2}_{\mathrm{loc}}$, and $|u^{(n)}|^2 u^{(n)} \to |u|^2 u$ in L^1_{loc}. Hence the terms with (Lemma 2.4) $(\partial_t\varphi+\Delta\varphi)$ and $u\cdot\nabla\varphi$ converge. For the pressure term, $p^{(n)} \rightharpoonup p$ in $L^{3/2}_{\mathrm{loc}}$ and $u^{(n)} \cdot \nabla\varphi \to u \cdot \nabla\varphi$ strongly in L^3_{loc}, so $p^{(n)}(u^{(n)} \cdot \nabla\varphi) \to p(u \cdot \nabla\varphi)$ in L^1 by Hölder (NFS.EST.2). Finally (2.4a) follows from weak convergence of $\nabla u^{(n)}$ in L^2_{loc} and the convexity of $|\cdot|^2$ with nonnegative weight φ. □

The remainder of the manuscript is internal: starting from Theorem 2.1, we run the pinned LRNS spine, derive the Gaussian budget, establish Gate/No-Theft, and conclude by Tsai rigidity.

0.4 Spine dependency index (reader map)

The proof spine is organized into the following named steps; each step points to the precise internal statement(s).

1. **Main claim. Theorem 1.1** (global regularity).
2. **Critical element extraction. Theorem 2.1 (CE(L^3))** and its outputs; see Appendix NFS.IMP.
3. **Flux sealing / tail control.** Lemmas **2.2**–**2.4** and Lemma **4.1**.
4. **Pinned normalization + width observable.** Lemmas **5.1**–**5.3**.
5. **Gaussian OU* budget.** Lemma **6.1**.
6. **Gate/No-Theft dichotomy and dissipation packet accounting.** Lemmas **7.1F**, **7.1H**, **7.2**, **7.3**.
7. **Rigidity via Tsai.** Lemmas **7.6E1**, **7.6F**, **7.6G**, and the isolated nontriviality lemma **7.6P**.

A complete statement index (with local dependency hints and usage counts) is recorded in Appendix **NFS.SDI**.

1. Leray similarity RG normalization (CXM canonical packaging)

1.1 Similarity variables and the RG orbit as time shifts

Fix a putative singular point $(x_\star, T^\star)$. Introduce Leray similarity variables

$$y = \frac{x - x_\star}{\sqrt{T^\star - t}}, \qquad \tau = -\log(T^\star - t),$$

and the Leray-rescaled fields

$$U(y,\tau) = \sqrt{T^\star - t}\, u(x,t), \qquad P(y,\tau) = (T^\star - t)\, p(x,t).$$

In these variables the dynamics are governed by the Leray-rescaled Navier–Stokes system (LRNS) (see §2.2).

The discrete "RG windows" are simply **unit time translates in** τ:

$$U_k(y,s) := U(y, k+s), \qquad P_k(y,s) := P(y, k+s), \qquad s \in [-1, 0].$$

This is equivalent to the fixed-frame discrete rescalings (the scale factor changes by $e^{-1/2}$ per unit of τ), but it is the natural packaging for the energy budget.

1.2 Gauge fixing: translation pinning (no optional normalizations)

The package (GKP3) gives **spatial tightness modulo translations**. We fix a translation gauge so that all windows are centered at the origin.

Lemma 3.1 (Translation pinning). Assume (GKP3): for every $\varepsilon > 0$ there exists $R(\varepsilon)$ such that for each k there is a center $y_k \in \mathbb{R}^3$ with

$$\int_{|y - y_k| > R(\varepsilon)} |U_k(y,0)|^3\, dy \le \varepsilon.$$

Then after defining the translated windows

$$\widetilde{U}_k(y,s) := U_k(y + y_k, s),$$

we have the pinned tightness estimate

$$\int_{|y| > R(\varepsilon)} |\widetilde{U}_k(y,0)|^3\, dy \le \varepsilon \quad \text{for all } k.$$

Proof. This is immediate from the change of variables $z = y + y_k$. □

From now on we rename $\widetilde{U}_k$ back to U_k and assume the translation gauge has been fixed.

Remark (polish). Earlier development versions also imposed optional "vorticity-amplitude" normalizations. Those are **not used** in the present single-spine proof and are intentionally omitted here to prevent circularity and notational drift.

2. Flux Sealing (Hard Analysis) — the key analytic lemma

2.1 The only flux notion used (cutoff-flux)

Let $\chi \in C_c^\infty(\mathbb{R}^3)$ be radial, $\chi \equiv 1$ on $|y| \le 1$, $\chi \equiv 0$ on $|y| \ge 2$, and set $\chi_R(y) := \chi(y/R)$.

Define the **Gaussian cutoff-flux** over a unit window:

$$\mathcal{F}_{G,R}[U,P](s) := \int_{s-1}^{s} \int_{\mathbb{R}^3} \left(\tfrac{1}{2}|U|^2 + P\right) U \cdot \nabla(\chi_R^2)\, G\, dy\, d\tau.$$

Definition (Flux Sealing). The orbit satisfies **Flux Sealing** if

$$\lim_{R\to\infty} \sup_k \sup_{s\in[-1,0]} \left|\mathcal{F}_{G,R}[U_k,P_k](s)\right| = 0.$$

Only $\mathcal{F}_{G,R}$ is used; no "surface flux" is introduced.

2.2 Pressure control (Calderón–Zygmund (NFS.EST.4); uniform in time)

For suitable solutions on $\mathbb{R}^3$,

$$P = \mathcal{R}_i \mathcal{R}_j (U_i U_j)$$

and hence for each fixed τ, [Gra14, Cor. 5.2.8]

$$\|P(\cdot,\tau)\|_{L^{3/2}} \le C_{CZ}\, \|U(\cdot,\tau)\|_{L^3}^2,$$

with universal C_{CZ}; see e.g. [Gra14, Cor. 5.2.8].

2.3 Lemma S3 (Flux Sealing from critical compactness + CZ)

Lemma 4.1 (Flux Sealing / No-Theft). Assume (GKP1)–(GKP3). Then Flux Sealing holds:

$$\lim_{R\to\infty} \sup_k \sup_{s\in[-1,0]} \left|\mathcal{F}_{G,R}[U_k,P_k](s)\right| = 0.$$

Proof (Annals-grade; no hidden steps). Fix k. Let $A_R := \{R \le |y| \le 2R\}$. Note $\nabla(\chi_R^2)$ is supported on A_R and $|\nabla(\chi_R^2)| \lesssim 1/R$. Also $0 < G \le 1$.

For any $s \in [-1,0]$,

$$|\mathcal{F}_{G,R}[U_k,P_k](s)| \le \int_{s-1}^{s} \int_{A_R} \left(\tfrac{1}{2}|U_k|^2 + |P_k|\right) |U_k|\, |\nabla(\chi_R^2)|\, G\, dy\, d\tau.$$

Thus by Hölder (NFS.EST.2) and §2.2,

$$\begin{aligned}|\mathcal{F}_{G,R}[U_k,P_k](s)| &\lesssim \frac{1}{R} \int_{s-1}^{s} \left(\|U_k(\tau)\|_{L^3(A_R)}^3 + \|P_k(\tau)\|_{L^{3/2}} \|U_k(\tau)\|_{L^3(A_R)}\right) d\tau \\ &\lesssim \frac{1}{R} \int_{s-1}^{s} \left(\|U_k(\tau)\|_{L^3(A_R)}^3 + M^2 \|U_k(\tau)\|_{L^3(A_R)}\right) d\tau.\end{aligned}$$

By tightness (GKP2), for every $\varepsilon > 0$ there exists R_ε such that for all $R \ge R_\varepsilon$, $\|U_k(\tau)\|_{L^3(A_R)} \le \|U_k(\tau)\|_{L^3(|y|>R)} \le \varepsilon$ uniformly in k, τ. Hence for $R \ge R_\varepsilon$, (Lemma 4.1)

$$\sup_k \sup_{s\in[-1,0]} |\mathcal{F}_{G,R}[U_k,P_k](s)| \lesssim \frac{1}{R}\left(\varepsilon^3 + M^2\varepsilon\right) \xrightarrow[R\to\infty]{} 0,$$

and since ε is arbitrary, the Flux Sealing limit holds. $\square$

Interpretation. Flux Sealing is the rigorous form of "No-Theft": energy cannot be imported from infinity because the critical element is tight.

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3. Critical Coercivity (Modulation Cost) — what is needed and what is proven

This section isolates exactly what is required to force the ω-limit set to be a singleton.

3.1 A quantitative width observable (Gaussian energy quantile)

Definition 5.1 (-breathing window). Fix parameters (θ, κ) in the normalized class and let $\mathcal{R}_{\theta,\kappa}(s)$ denote the log-averaged quantile radius defined by (5.2). A normalized RG window U on $[-1, 0]$ is **σ-breathing** if

$$e^{-\sigma} \;\le\; \mathcal{R}_{\theta,\kappa}(s) \;\le\; e^{\sigma} \qquad \text{for all } s \in [-1, 0]. \tag{5.1}$$

(Equivalently, the instantaneous profile does not concentrate or spread beyond a fixed multiplicative factor on the unit window.)

We measure the "width" of a window by **where the Gaussian energy profile crosses a band of levels**. The key point is that we must avoid any hidden "crossing slope" assumption: if the profile is nearly flat near a single threshold, then an $\|F - G\|_{L^\infty_R}$ perturbation can create an arbitrarily large hitting-time drift. We therefore define the width by **averaging quantile radii across a hysteresis band** (Appendix NFS.IMP).

For a window (U_k, P_k) and a slice $s \in [-1, 0]$ define the *Gaussian energies*

$$E_R(s) := \int_{\mathbb{R}^3} \tfrac{1}{2}|U_k(y, s)|^2\, \chi_R(y)^2\, G(y)\, dy, \qquad E_\infty(s) := \int_{\mathbb{R}^3} \tfrac{1}{2}|U_k(y, s)|^2\, G(y)\, dy,$$

and the *instantaneous normalized profile*

$$\widetilde{F}_s(R) := \frac{E_R(s)}{E_\infty(s)} \in [0, 1], \qquad R \ge 1. \tag{5.0}$$

(If $E_\infty(s) = 0$ then $U_k(\cdot, s) = 0$ a.e. and the window is trivial; in the canonical spine we work on nontrivial windows, so $E_\infty(s) > 0$ and $\widetilde{F}_s$ is well-defined.)

Quantile radii. For $\alpha \in (0, 1)$ define the *quantile radius*

$$Q_s(\alpha) := \inf\{R \ge 1 : \; \widetilde{F}_s(R) \ge \alpha\}. \tag{5.1}$$

Since $R \mapsto \widetilde{F}_s(R)$ is monotone increasing and $\widetilde{F}_s(R) \uparrow 1$ as $R \to \infty$, each $Q_s(\alpha)$ is finite.

quantitative width radius (Option B). Fix $\theta \in (0, 1)$ and a band half-width $\kappa \in (0, \theta \wedge (1-\theta))$. We define the (θ, κ)–width radius at slice s by the *logarithmic band-average* of quantiles:

$$\log \mathcal{R}_{\theta,\kappa}(s) := \frac{1}{2\kappa} \int_{\theta-\kappa}^{\theta+\kappa} \log Q_s(\alpha)\, d\alpha. \tag{5.2}$$

Equivalently, $\mathcal{R}_{\theta,\kappa}(s)$ is the geometric mean of the quantile radii across the band $[\theta-\kappa, \theta+\kappa]$.

Admissible band. We fix parameters $0 < \theta < 1$ and $0 < \kappa < \min\{\theta, 1-\theta\}$ so that the band $[\theta-\kappa, \theta+\kappa] \subset (0,1)$.

This definition is **hysteretic by construction**: it is stable under L^∞_R perturbations of the profile without any "non-flatness at the crossing" assumption. The deterministic drift conversion is proved in §4.2.6.5.

Window width. We evaluate (5.2) at the *pin time* $s = 0$ and write

$$\mathcal{R}_k := \mathcal{R}_{\theta,\kappa}[U_k](0). \tag{5.3}$$

The "breathing" alternative of M1 concerns whether $\mathcal{R}_k$ stays in a fixed compact interval or exhibits multiplicative expansions/contractions.

3.2 Non-degeneracy: pinned gauge forces positive Gaussian energy

Lemma 5.1 (Gaussian energy non-degeneracy at the pin time). Assume (GKP1)–(GKP4). Then there exists $m_0 = m_0(R_{\mathrm{pin}}, \eta_{\mathrm{pin}}) > 0$ such that

$$\inf_k E_G[U_k](0) \;\geq\; m_0. \tag{5.1a}$$

Moreover, once the full-space Leray–Gaussian budget of §4 is established (so that $E_G(\tau)$ is non-increasing in τ), one also has

$$\inf_k \inf_{s\in[-1,0]} E_G[U_k](s) \;\geq\; m_0. \tag{5.1b}$$

Proof. Fix k. By (GKP4), $\|U_k(\cdot,0)\|_{L^3(B_{R_{\mathrm{pin}}})} \geq \eta_{\mathrm{pin}}$. By Hölder on the finite-measure set $B_{R_{\mathrm{pin}}}$, $\|U_k(\cdot,0)\|_{L^2(B_{R_{\mathrm{pin}}})} \;\geq\; \|U_k(\cdot,0)\|_{L^3(B_{R_{\mathrm{pin}}})}\,|B_{R_{\mathrm{pin}}}|^{\frac{1}{6}} \;\geq\; \eta_{\mathrm{pin}}\,|B_{R_{\mathrm{pin}}}|^{\frac{1}{6}}$. Squaring and using that $G(y) \geq g_0 := (4\pi)^{-3/2}e^{-R_{\mathrm{pin}}^2/4}$ on $B_{R_{\mathrm{pin}}}$, $E_G[U_k](0) = \int_{\mathbb{R}^3} \frac{1}{2}|U_k(y,0)|^2 G(y)\,dy \;\geq\; \frac{1}{2}\,g_0 \int_{B_{R_{\mathrm{pin}}}} |U_k(y,0)|^2\,dy \;\geq\; \frac{1}{2}\,g_0\,\eta_{\mathrm{pin}}^2\,|B_{R_{\mathrm{pin}}}|^{\frac{1}{3}}$. Define $m_0 := \frac{1}{2}\,(4\pi)^{-3/2}e^{-R_{\mathrm{pin}}^2/4}\,\eta_{\mathrm{pin}}^2\,|B_{R_{\mathrm{pin}}}|^{\frac{1}{3}} > 0$. This proves (5.1a).

For (5.1b), once §4 yields the full-space inequality $E_G(\tau_2) + \int_{\tau_1}^{\tau_2} D_G(\tau)\, d\tau \leq E_G(\tau_1)$ for all $\tau_1 < \tau_2$, E_G is non-increasing in τ. Hence on each window $\tau \in [k-1, k]$ (i.e. $s \in [-1, 0]$), (Lemma 5.1) $E_G[U_k](s) = E_G(k+s) \geq E_G(k) = E_G[U_k](0) \geq m_0$, so (5.1b) follows. $\square$

Lemma 5.2 (Singleton ω-limit under coercive modulation cost). Let $\{U(s)\}_{s\leq 0}$ be a precompact trajectory in a metric space $(\mathcal{K}, d)$. Assume there exists a nonnegative "modulation cost" $m(s) \in L^1((-\infty, 0])$ and a constant $C > 0$ such that

$$d\big(U(s_1), U(s_2)\big) \;\le\; C \int_{s_1}^{s_2} m(s)\, ds. \tag{5.2}$$

Then $U(s)$ has a unique limit in $\mathcal{K}$ as $s \to -\infty$; equivalently, the ω-limit set is a singleton.

Proof. The bound (5.2) implies $\{U(s)\}_{s\leq 0}$ is a Cauchy family as $s \to -\infty$ because $\int_{-\infty}^{0} m(s)\, ds < \infty$. Since the trajectory is precompact, the Cauchy criterion yields convergence in $\mathcal{K}$. $\square$

Comment. The lower bound is driven by the *non-evanescence* pinning (GKP4), not by pointwise vorticity continuity. This is deliberate: it makes the non-degeneracy statement stable in the suitable-solution setting and avoids hidden regularity assumptions.

Lemma 5.1A (Quantile confinement on non-degenerate slices; proof-fix). Fix (θ, κ) as in (5.2) and set $\alpha_- := \theta - \kappa \in (0, 1)$. Assume a slice $s \in [-1, 0]$ satisfies

$$\|U_k(\cdot,s)\|_{L^3(\mathbb{R}^3)} \le M_3, \qquad E_\infty(s) \ge m_1 > 0. \tag{5.4}$$

Then there exists $R_+ = R_+(M_3, m_1, \alpha_-)$ such that for all $u \in [\alpha_-, \theta + \kappa]$,

$$1 \le Q_s(\alpha) \le R_+, \qquad \text{hence} \qquad (Lemma 5.1A) 1 \le \mathcal{R}_{\theta,\kappa}(s) \le R_+. \tag{5.5}$$

Proof. Since $\chi_R \equiv 1$ on B_R and $0 \le \chi_R \le 1$, $E_\infty(s) - E_R(s) = \int \frac{1}{2}|U_k|^2(1-\chi_R^2)G \le \frac{1}{2}\int_{|y|\ge R} |U_k(y,s)|^2\, G(y)\, dy$. By Hölder with exponents $(3/2, 3)$, $\int_{|y|\ge R} |U_k|^2 G \le \left(\int_{\mathbb{R}^3} |U_k(y,s)|^3\, dy\right)^{2/3} \left(\int_{|y|\ge R} G(y)^3\, dy\right)^{1/3} \le M_3^2 \left(\int_{|y|\ge R} G^3\right)^{1/3}$. The Gaussian tail satisfies $\int_{|y|\ge R} G(y)^3\, dy \le C\, e^{-3R^2/4}$ for $R \ge 1$. Choose R_+ so that $\frac{1}{2} M_3^2 \left(\int_{|y|\ge R_+} G^3\right)^{1/3} \le (1-\alpha_-)\, m_1$. Then for every $R \ge$

Remark. In the proof spine we will apply Lemma 5.1A at $s = 0$ with $m_1 = m_0$ (Lemma 5.1) and at $s = -1$ with $m_1 \simeq m_0$ once the pinned profile stability (7.14pin) is available (see §4.2.6.5). No full-space monotonicity of E_G is used.

3.3 The modulation-cost statement needed downstream (M1 closed)

We require a **uniform** positive dissipation cost for any genuine "breath" of the critical profile over one fixed RG window.

Lemma I.4 (Modulation cost / Critical coercivity). Fix $\sigma \in (0,1)$. There exists $\varepsilon_*(\sigma) > 0$ (depending only on σ and the normalized-class bounds (CE constants; cf. Appendix NFS.1.9(B))) such that for every normalized RG profile U on $[-1,0]$,

- if U is σ-breathing on $[-1, 0]$ in the sense of Definition 5.1, then $\int_{-1}^{0} D_G[U](s)\,ds \ \geq \ \varepsilon_*(\sigma)$.

Proof. We argue by **compactness/contradiction**, which automatically yields a constant that is **uniform** over the entire normalized class.

Step 1 (Set-up: the compact window class). Let $\mathcal{K}$ denote the collection of normalized RG windows U on $[-1, 0]$ that satisfy:

1. U is a suitable weak solution of the Leray-rescaled Navier–Stokes system (LRNS) on $[-1,0]\times\mathbb{R}^3$. 2. Uniform critical bound: $\|U\|_{L^\infty([-1,0];L^3(\mathbb{R}^3))}\leq M$. 3. Tightness (GKP): for every $\eta>0$ there exists $R(\eta)$ such that $\sup_{s\in[-1,0]}\int_{|y|>R(\eta)}|U(y,s)|^3\,dy\leq\eta$. 4. Non-evanescence at time 0: $E_G[U](0)\geq m_0>0$ (Lemma 5.1 (in-equal-

(Only (2)–(4) are used in this proof; the LRNS structure is used only to justify the compactness statement below.)

Step 2 (Compactness and lower semicontinuity). Assume for contradiction that no uniform $\varepsilon_*(\sigma)$ exists. Then there is a sequence $U^{(n)} \in \mathcal{K}$ that is σ-breathing on $[-1,0]$ and $\int_{-1}^{0} D_G[U^{(n)}](s)\,ds \to 0$. Since G is bounded below on every fixed ball B_R, this

Now use the global critical bound and tightness: for each such s, $U^{(\infty)}(s,\cdot) \in L^3(\mathbb{R}^3)$. The only spatially constant L^3 field is 0. Therefore (0) $U^{(\infty)} \equiv 0 \quad \text{on } [-1,0]\times\mathbb{R}^3$, and in particular $E_G[U^{(\infty)}](0) = 0$.

Step 3 (Contradiction with non-evanescence and closedness of the window topology). The strong convergence on every ball, together with tightness, implies convergence at time 0 in the weighted space: $U^{(n)}(0) \to U^{(\infty)}(0)$ strongly in $L^2(G\,dy)$. (Indeed: split $\mathbb{R}^3$ into $B_R \cup B_R^c$, pass

By Lemma 5.1 (inequality (5.1a)), $E_G[U^{(n)}](0) \geq m_0$ for all n. Passing to the limit yields $E_G[U^{(\infty)}](0) \geq m_0$, contradicting $E_G[U^{(\infty)}](0) = 0$. This contradiction shows that $\varepsilon_*(\sigma) := \inf\Big\{ \int_{-1}^{0} D_G[U](s)\,ds : U \in \mathcal{K} \text{ is } \sigma\text{-breathing}\Big\} > 0$, which proves the lemma. $\square$

Remark 5.5 (What M1 was, and why it is now closed). Earlier drafts tried to force a *quantitative* bound on the boundary-flux term at the (possibly $O(1)$) "width radius". The argument above avoids that entirely: we only use (i) compactness of the nor-

4. Energy Budget in Leray Similarity Variables (OU–Adjoint Form)

4.1 Why we change frames

In earlier drafts of this spine the Gaussian-weighted local energy inequality (LEI) was written directly for the fixed-frame RG windows

$$U_k(x,t) = r_k\, u(x_k + r_k x, t_k + r_k^2 t), \qquad t \in [-1,0],$$

with the fixed backward heat kernel $G(x) = (4\pi)^{-3/2} e^{-|x|^2/4}$. The resulting term

$$B_G[k] = \iint_{[-1,0]\times\mathbb{R}^3} \Big(\tfrac{1}{2}|U_k|^2 \Delta G + (\tfrac{1}{2}|U_k|^2 + P_k) U_k \cdot \nabla G \Big)\, dx\, dt$$

has no *a priori* sign and (as written) is not obviously summable in k.

The concentration–compactness (see [GKP13]) route (NRS96) passes to **Leray similarity variables**, where "zooming-in" is encoded as *time translation*. In that frame the Gaussian G becomes the natural stationary density for the adjoint of the linearized generator, and the problematic "bulk" terms collapse to annulus-supported errors after a cutoff.

This section implements that change and reduces the previous P1 ("bulk term") to a single explicit verification item (VI.6) plus the already-defined Flux Sealing.

4.2 Leray similarity variables and the rescaled equation

Fix a putative singular point $(x_\star, T^\star)$ and set similarity variables

$$y = \frac{x - x_\star}{\sqrt{T^\star - t}}, \qquad \tau = -\log(T^\star - t),$$

and the Leray-rescaled fields

$$U(y,\tau) = \sqrt{T^\star - t}\; u(x,t), \qquad P(y,\tau) = (T^\star - t)\; p(x,t).$$

A direct chain-rule computation gives the **Leray-rescaled Navier–Stokes system**

$$\partial_\tau U - \Delta U + (U\cdot\nabla)U + \nabla P + \tfrac{1}{2}\, y\cdot\nabla U + \tfrac{1}{2}\, U = 0, \qquad \nabla\cdot U = 0. \tag{LRNS}$$

(For the stationary/self-similar case $U(\cdot,\tau) \equiv U(\cdot)$ this reduces to the profile equation $-\Delta U + \frac{1}{2}\, y\cdot\nabla U + \frac{1}{2}\, U + (U\cdot\nabla)U + \nabla P = 0$, consistent with Tsai's formulation. (see Tsai98).)

Discrete RG orbit as time shifts. One may now define the RG windows by **translation in τ**:

$$U_k(y,s) := U(y, k+s), \qquad s \in [-1,0],$$

so that the former "scale-to-scale" step becomes the trivial map $k \mapsto k+1$.

Pinned gauge in the Leray frame. The earlier "pinned vorticity" normalization can be imposed at each integer time $\tau = k$ by composing with spatial translations in y (and, if needed, rotations), exactly as in the fixed-frame orbit. This is the same modulation content, just expressed in the natural similarity time.

4.3 The OU–adjoint identity for the Gaussian

Let

$$G(y) = (4\pi)^{-3/2} \exp\Big(- \frac{|y|^2}{4} \Big).$$

Then

$$\nabla G = -\frac{y}{2} G, \qquad \Delta G = \Big(\frac{|y|^2}{4} - \frac{3}{2} \Big) G,$$

and the crucial **OU–adjoint relation**

$$\Delta G + \frac{1}{2}\, y \cdot \nabla G + \frac{3}{2}\, G = 0 \tag{OU*}$$

holds identically (a one-line substitution).

This identity is the reason we pass to (LRNS): the additional linear terms $-\frac{1}{2} y \cdot \nabla U - \frac{1}{2} U$ in (LRNS) are precisely the terms whose adjoint produces the $\frac{1}{2} y \cdot \nabla \psi + \frac{3}{2} \psi$ correction in the LEI; choosing ψ proportional to G annihilates the bulk contribution.

4.4 Localized Gaussian energy inequality in the Leray frame (VI.6 filled)

We record the localized Gaussian energy inequality in similarity variables, and we fill in the only missing calculus check from the previous draft (the origin of the "+3" coefficient).

Lemma 6.1 (Localized Gaussian budget in the Leray frame). Let (U, P) be a suitable weak solution to (LRNS) on $[-1, 0] \times \mathbb{R}^3$. Fix $R \geq 1$ and set

$$\psi_R(y) := \chi_R(y)^2\, G(y), \qquad E_R(s) := \int \tfrac{1}{2} |U(y,s)|^2\, \psi_R(y)\, dy, \qquad D_R(s) := \int |\nabla U(y,s)|^2\, \psi_R(y)\, dy.$$

Then for all $-1 \leq s_1 < s_2 \leq 0$,

$$E_R(s_2) + \int_{s_1}^{s_2} D_R(s)\, ds \;\leq\; E_R(s_1) + \mathrm{Err}_R[U; s_1, s_2], \tag{6.2}$$

(General form.) Lemma 6.1 is a special case of the LRNS local energy inequality (LRNS-LEI) recorded in Appendix **NFS.1.5**.

where

$$\mathrm{Err}_R[U; s_1, s_2] := \int_{s_1}^{s_2} \int \Big(\tfrac{1}{2} |U|^2 + P \Big)\, U \cdot \nabla \psi_R\, dy\, ds + \int_{s_1}^{s_2} \int \Big[\tfrac{1}{2} |U|^2\, \Delta \psi_R + \tfrac{1}{4} |U|^2\, (y \cdot \nabla \psi_R + 3 \psi_R) \Big] dy\, ds.$$

Proof. Start from the local energy inequality for (u, p) in physical variables and test it with a localized backward heat kernel:

$$\phi_R(x,t) := \chi_R\left(\frac{x}{\sqrt{s}} \right)^2 \Gamma(x,t) \quad \text{with} \quad s := t_* - t > 0, \qquad \Gamma(x,t) = (4\pi s)^{-3/2} e^{-|x|^2/(4s)}.$$

Introduce similarity variables

$$y := \frac{x}{\sqrt{s}}, \qquad \tau := -\log s, \qquad u(x,t) = \frac{1}{\sqrt{s}} U(y,\tau), \qquad p(x,t) = \frac{1}{s} P(y,\tau),$$

so that (U,P) solves (LRNS). In these variables one has $\Gamma(x,t) = s^{-3/2} G(y)$ with $G(y) = (4\pi)^{-3/2} e^{-|y|^2/4}$, and therefore $\phi_R(x,t) = s^{-3/2}\psi_R(y)$ (Appendix NFS.IMP).

The LEI term $(\partial_t + \Delta_x)\phi_R$ produces the "OU*" operator. The missing check (VI.6) is the following computation:

(VI.6) Similarity calculus for $(\partial_t + \Delta_x)\phi_R$. Writing $s = t_* - t$ and $\phi_R = s^{-3/2}\psi_R(y)$ with $y = x/\sqrt{s}$, one has

$$(\partial_t + \Delta_x)\phi_R = \frac{1}{2}\, s^{-5/2}\, (2\Delta_y + y \cdot \nabla_y + 3)\psi_R(y).$$

Proof. Since $\partial_t = -\partial_s$, one computes

$$\partial_s(s^{-3/2}\psi_R(y)) = -\tfrac{3}{2} s^{-5/2}\psi_R(y) - \tfrac{1}{2} s^{-5/2}\, y \cdot \nabla\psi_R(y),$$

while $\Delta_x(s^{-3/2}\psi_R(y)) = s^{-5/2}\Delta_y\psi_R(y)$. Adding gives

$$(\partial_t + \Delta_x)\phi_R = s^{-5/2}\Big[\Delta\psi_R + \tfrac{1}{2} y \cdot \nabla\psi_R + \tfrac{3}{2}\psi_R\Big] = \tfrac{1}{2} s^{-5/2}(2\Delta + y \cdot \nabla + 3)\psi_R.$$

The coefficient "+3" comes *only* from differentiating the prefactor $s^{-3/2}$.

$$(\partial_t + \Delta_x)\phi_R = \tfrac{1}{2}\, s^{-5/2}\, (2\Delta_y + y \cdot \nabla_y + 3)\psi_R(y).$$

In particular, the factor is $(\frac{1}{2})$, not $(\frac{1}{4})$.

Substituting this identity into the LEI (and rewriting the remaining LEI terms under the same change of variables) yields (6.2). □

Remark 6.2 (OU* localization for $\psi_R = \chi_R^2 G$). The adjoint OU identity $(2\Delta + y \cdot \nabla + 3)G = 0$ implies

$$(2\Delta + y \cdot \nabla + 3)(\chi_R^2 G) = (2\Delta\chi_R^2 + y \cdot \nabla\chi_R^2)G + 4\nabla\chi_R^2 \cdot \nabla G,$$

so the entire $(\partial_t + \Delta)$-forcing generated by the cutoff is supported on the annulus $A_R = \{R \le |y| \le 2R\}$.

4.5 Passing $R \to \infty$: what survives, and the correct full-space Gaussian identity (P1_G)

Lemma 6.1 is the *only* legitimate bridge between the physical local-energy inequality and any Gaussian-weighted budget in the Leray-rescaled variables. A crucial subtlety is that the error term $\mathrm{Err}_R[U_k]$ contains a **bulk** contribution coming from the factor $\chi_R^2 \nabla G$:

$$U_k \cdot \nabla\psi_R = U_k \cdot (\chi_R^2 \nabla G) \;+\; U_k \cdot (G\nabla(\chi_R^2)).$$

The second term is annular (supported in A_R), but the first term is *not*: it persists on $\{|y| \lesssim R\}$ and converges to $U_k \cdot \nabla G$ as $R \to \infty$.

This means:

- The *forcing* part of Err_R becomes annular after expanding $\psi_R = \chi_R^2 G$ (because $(\Delta + \frac{1}{2} y \cdot \nabla + \frac{3}{2})G = 0$), hence it vanishes as $R \to \infty$ under (GKP) + cutoff bounds (NFS.EST.1).

- The *annular* transport pieces (those involving $\nabla\chi_R$) also vanish as $R \to \infty$ under (GKP) and Flux Sealing.
- But the **bulk Gaussian transport flux**

$$\mathcal{F}_G[U_k; s_1, s_2] := \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\tfrac{1}{2}|U_k|^2 + P_k\right) U_k \cdot \nabla G \, dy \, ds$$

  survives in the limit and has **no sign a priori**.

So the "monotone full-space Gaussian budget" claimed in earlier drafts is not available unless one proves an additional cancellation/sign statement for $\mathcal{F}_G$. We elevate that missing input to an explicit gap.

Proposition 6.3 (Correct full-space Gaussian LEI in LRNS). Let (U_k, P_k) be a suitable LRNS solution on $[-1, 0]$ as in §3. Then for all $-1 \le s_1 < s_2 \le 0$,

$$E_G[U_k](s_2) \; + \; \int_{s_1}^{s_2} D_G[U_k](s) \, ds \; \le \; E_G[U_k](s_1) \; + \; \mathcal{F}_G[U_k; s_1, s_2], \tag{6.3}$$

where

$$E_G[U_k](s) := \int_{\mathbb{R}^3} \tfrac{1}{2}|U_k(y,s)|^2 \, G(y) \, dy, \qquad D_G[U_k](s) := \int_{\mathbb{R}^3} |\nabla U_k(y,s)|^2 \, G(y) \, dy.$$

Proof. Apply Lemma 6.1 with $\psi_R = \chi_R^2 G$ and then let $R \to \infty$.

1. The left-hand side converges to $E_G[U_k](s_2) + \int_{s_1}^{s_2} D_G[U_k]$ by monotone convergence, since $\psi_R \uparrow G$ and $|\nabla U_k|^2 \psi_R \uparrow |\nabla U_k|^2 G$.
2. For the forcing part of Err_R, expand $(\Delta + \frac{1}{2} y \cdot \nabla + \frac{3}{2})(\chi_R^2 G)$ and use $(\Delta + \frac{1}{2} y \cdot \nabla + \frac{3}{2}) G = 0$. Every remaining term contains $\nabla\chi_R$ or $\Delta\chi_R$, hence is supported in A_R and tends to 0 as (Proposition 6.3) $R \to \infty$ by the (GKP) tightness bounds.
3. For the transport part,

$$\int_{s_1}^{s_2} \int \left(\tfrac{1}{2}|U_k|^2 + P_k\right) U_k \cdot \nabla(\chi_R^2 G) = \mathcal{F}_G[U_k; s_1, s_2] \; + \; o_{R\to\infty}(1),$$

   where the $o(1)$ comes from the annular $\nabla\chi_R$ contribution, controlled exactly as in step 2 together with Flux Sealing to handle the pressure terms. This yields (6.3).

P1_G (Gaussian flux sign/cancellation). Prove that along the pinned RG orbit (or for every LRNS suitable solution in the GKP class), the flux term $\mathcal{F}_G[U_k; s_1, s_2]$ is nonpositive, or at least summable in k in a way that implies a finite total dissipation bound. Without such an input, (6.3) does **not** imply that E_G is monotone.

Remark (how we avoid using P1_G in the proof spine). The contradiction argument in §6 does *not* require monotonicity of the **full-space** E_G. What it needs is a finite-dissipation budget at the **quantitative scale** $\mathcal{R}_k$. That budget is obtained by keeping cutoffs (and eventually allowing them to *move*) and then using the M2 trade mechanism to absorb the cutoff-generated flux into the dissipation. The rest of §6 is written so that P1_G becomes optional: it would simplify some steps, but it is not used as a external step.

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5. Gate and No-Theft mechanisms (integration targets)

Gate definition

We use the **Gaussian shell cubic flux gate**

$$\mathcal{G}(R,\tau) := \frac{1}{R}\int_{A_R} |U(\tau)|^3\, G\, dy, \qquad A_R := \{R \le |y| \le 2R\}.$$

A "gate opening" event is the existence of (R,τ) such that $\mathcal{G}(R,\tau) \ge \gamma_0$.

No-Theft principle (pressure-tail decoupling)

The pressure is slaved: $P = \mathcal{R}_i\mathcal{R}_j(U_iU_j)$.

Definition (No-Theft tail). For an LRNS profile U on $[-1,0]$, define the pressure-tail functional

$$\mathrm{Tail}_\sharp(R,\tau) := \frac{1}{R^3}\int_{B_{R/8}} |U(y,\tau)|^2\, dy \;+\; \int_{|y|\ge 16R} \frac{|U(y,\tau)|^2}{|y|^3}\, dy, \qquad R \ge 1,\ \tau \in [-1,0]. \tag{7.NT0}$$

It measures the far-field contribution of the Calderón–Zygmund pressure kernel to the core region after the near/far split.

Lemma 7.8C$_{\mathrm{NT}}$ (No-Theft tail decay). For every $\tau \in [-1,0]$ and every $R \ge 1$,

$$\mathrm{Tail}_\sharp(R,\tau) \;\le\; \frac{C_{\mathrm{NT}}}{R^2}\, \|U(\tau)\|^2_{L^3(\mathbb{R}^3)}, \tag{7.NT1}$$

with an absolute constant C_{NT} depending only on the fixed cutoff template. Under (CE1), $\sup_{\tau\in[-1,0]} \mathrm{Tail}_\sharp(R,\tau) \lesssim M^2R^{-2}$, hence the tail remainder is uniform in time and vanishes as $R \to \infty$ (Lemma 7.8C).

Proof. Hölder on $B_{R/8}$ gives $\int_{B_{R/8}} |U|^2 \le \|U\|_3^2 |B_{R/8}|^{1/3}$, hence the first term $\lesssim R^{-2}\|U\|_3^2$ (Lemma 7.8C). For the second term, use $\||U|^2\|_{3/2} = \|U\|_3^2$ and $\||y|^{-3}\|_{L^3(|y|\ge 16R)} \sim R^{-2}$. □

We introduce $\mathrm{Tail}_\sharp(R,\tau)$ and prove **uniform-in-time** vanishing as $R \to \infty$. This ensures the far-field cannot inject order-one flux into the core on the blow-up time scale.

Flux Sealing as the "energy barrier"

The Gate functional is the Gaussian-shell cubic flux

$$\mathcal{G}(R,\tau) := \frac{1}{R}\int_{A_R} |U(\tau)|^3\, G\, dy, \qquad A_R := \{R \le |y| \le 2R\}.$$

Lemma 7.8, Step 3 provides the pointwise estimate (cf. (7.20C11)):

$$\mathcal{G}(R,\tau) \;\le\; C\, M\, D_G[U](\tau; 4R), \qquad \text{for a.e. } \tau \in [-1,0],\ R \ge 1, \tag{7.G1}$$

where $D_G[U](\tau;4R) = \int_{\mathbb{R}^3} |\nabla U(\tau)|^2\, \chi_{4R}^2\, G\, dy$.

Lemma 7.8, Step 4 provides pressure localization (cf. (7.20C13)–(7.20C16)):

$$\frac{1}{R}\int_{A_R} |U|\,|P-(P)_{B_{2R}}|\,G\,dy \;\le\; C\,\frac{1}{R}\int_{A^{8R}_{R/4}} |U|^3\,G\,dy \;+\; C\,D_G[U](\tau;4R)^{1/2}\,\mathrm{Tail}_\sharp(R,\tau). \tag{7.G2}$$

By Young's inequality, the second term is $\le \varepsilon D_G[U](\tau;4R) + C_\varepsilon\,\mathrm{Tail}_\sharp(R,\tau)^2$. Uniform No-Theft (Lemma 7.8$\mathrm{C_{NT}}$) ensures that, along each RG window, the tail remainder $\sup_{\tau\in[-1,0]}\mathrm{Tail}_\sharp(R,\tau)^2$ is both **uniform** in time and vanishes as $R\to\infty$.

We record the barrier implication in the precise form actually used downstream.

Proposition 7.G (Gate barrier; "opening" forces dissipation cost). Let (U,P) be an RG-window profile on $[-1,0]$ satisfying (GKP1–GKP4). Fix $R\ge 1$ and an interval $I=[\tau_0,\tau_1]\subset[-1,0]$. If $\mathcal{G}(R,\tau)\ge\gamma$ for a.e. $\tau\in I$, then

$$\int_{\tau_0}^{\tau_1} D_G[U](\tau;4R)\,d\tau \;\ge\; \frac{\gamma}{C\,M}\,(\tau_1-\tau_0), \tag{7.G3}$$

with the same constant C as in (7.G1). Moreover, combining the fixed-cutoff localized energy identity (Lemma 7.10) with the strong trade (Lemma 7.8) yields

$$\big|E_G[U](\tau_1;R)-E_G[U](\tau_0;R)\big| \;\le\; C\int_{\tau_0}^{\tau_1} D_G[U](\tau;4R)\,d\tau \;+\; C\Big(\sup_{\tau\in I}\mathrm{Tail}_\sharp(\tau;R/4)\Big)^2. \tag{7.G4}$$

In particular, on large scales $R\gg 1$ (where No-Theft makes the tail term negligible), a sustained gate opening on any interval forces an order-one variation of the localized Gaussian energy.

Remark (what still must be explicit to convert this into the "pointwise Gate Theorem"). To deduce $\operatorname{ess\,sup}_{\tau\le 0,R\ge 1}\mathcal{G}(R,\tau)\le\gamma_0$ from (7.G3)–(7.G4), one needs a **persistence/anti-flash lemma** preventing $\mathcal{G}(R,\tau)$ from spiking on a null set of times. We isolate this as a single verification item.

Lemma 7.GA (Anti-Flash at the fixed gate threshold). Fix a gate level $\gamma>0$. Assume the CE(L^3) package CE1–CE4 holds on an RG window $\tau\in[-1,0]$. Then for every $R\ge 1$ the shell-gate functional

$$\mathcal{G}(R,\tau)=\frac{1}{R}\int_{A_R}|U(\tau)|^3\,G\,dy$$

admits a representative that is **continuous in** τ on $[-1,0]$. Consequently, there exists a time-length $c_\star=c_\star(M,\gamma)\in(0,1)$ such that whenever (Lemma 7.GA) $\mathcal{G}(R,\tau_\star)\ge 2\gamma$ at some $\tau_\star\in[-1,0]$, one has

$$\mathcal{G}(R,\tau)\ge\gamma \qquad \text{for all } \tau\in[\tau_\star,\min\{\tau_\star+c_\star,0\}]. \tag{7.G5}$$

In particular, a pointwise gate opening at level 2γ forces a **nontrivial persistence interval** at level γ.

Proof (reader-facing, "why continuity holds"). The only input here is the *endpoint critical regularity mechanism* already implicit in CE1–CE3.

1) **From CE1 to smoothness on the window.** By CE1 we have $U \in L^\infty_\tau L^3_y(\mathbb{R}^3)$ on $[-1,0]$, and by CE3 the pair (U,P) is suitable. The endpoint Serrin criterion / backward-uniqueness theorem (Escauriaza–Seregin–Šverák) implies that any suitable (Lemma 7.GA) Navier–Stokes solution with $L^\infty_t L^3_x$ control is regular on the corresponding cylinder. After translating between physical variables and similarity variables (which preserves the L^3 norm), we obtain that U is smooth on $(-1,0] \times \mathbb{R}^3$, hence in particular (Lemma 7.GA)

$$U \in C^\infty_{(\tau,y)}((-1,0] \times \mathbb{R}^3).$$

(See Appendix NFS, "Citation check 7.GA", for the cite-level mapping.)

2) **Continuity of the gate functional.** Fix $R \geq 1$. Since $U(\tau,\cdot)$ is smooth and $|U(\tau)|^3 G \in L^1(\mathbb{R}^3)$ for each $\tau \in [-1,0]$ (using CE1 and $0 < G \leq 1$), the map

$$\tau \longmapsto \int_{A_R} |U(\tau)|^3 \, G \, dy$$

is continuous on $[-1,0]$ by dominated convergence (or directly by smooth dependence of U on τ). Therefore $\tau \mapsto \mathcal{G}(R,\tau)$ is continuous on $[-1,0]$ (Lemma 7.GA).

3) **Uniform continuity and persistence at a fixed level.** A continuous function on the compact interval $[-1,0]$ is uniformly continuous. Thus, for the *fixed* level $\gamma > 0$ there exists $c_\star = c_\star(M,\gamma) \in (0,1)$ such that (Lemma 7.GA) $|\tau - \tau'| \leq c_\star$ implies $|\mathcal{G}(R,\tau) - \mathcal{G}(R,\tau')| \leq \gamma$. If $\mathcal{G}(R,\tau_\star) \geq 2\gamma$, this yields $\mathcal{G}(R,\tau) \geq \gamma$ for all $\tau \in [\tau_\star, \tau_\star + c_\star] \cap [-1,0]$, proving (7.G5).

Remark (why $c_\star$ is an admissible fixed constant in the Gate Theorem). In the Gate Theorem we apply Lemma 7.GA only at a **single, fixed threshold** $\gamma = \gamma_0$. Hence $c_\star(M,\gamma_0)$ is a fixed admissible constant for the entire argument (in the sense of Appendix NFS.1.9) (Lemma 7.GA). Moreover, for very large radii R the Gaussian factor makes $\mathcal{G}(R,\tau)$ automatically tiny:

$$\mathcal{G}(R,\tau) \leq \frac{1}{R} \sup_{A_R} G \int_{\mathbb{R}^3} |U(\tau)|^3 \, dy \leq \frac{M^3}{R} \, (4\pi)^{-3/2} e^{-R^2/4}.$$

Thus any potential gate opening occurs only for R in a compact range depending on (M,γ_0), (Lemma 7.GA) which is the natural regime for a uniform persistence constant.

Corollary 7.GB (A gate opening contains a fixed dissipation packet). Fix the threshold $\gamma = \gamma_0$ and let $c_\star = c_\star(M,\gamma_0)$ be given by Lemma 7.GA. If $\mathcal{G}(R,\tau_\star) \geq 2\gamma_0$ for some $R \geq 1$ and $\tau_\star \in [-1,0]$, then

$$\int_{\tau_\star}^{\min\{\tau_\star + c_\star, 0\}} D_G[U](\tau; 4R) \, d\tau \;\geq\; \frac{\gamma_0}{C\,M} \, c_\star, \tag{7.G6}$$

where C is the same constant as in (7.G1).

Proof. By Lemma 7.GA (with $\gamma = \gamma_0$), $\mathcal{G}(R,\tau) \geq \gamma_0$ for all $\tau \in [\tau_\star, \tau_\star + c_\star] \cap [-1,0]$. On this interval, (7.G1) implies $D_G[U](\tau; 4R) \geq \gamma_0/(C\,M)$ pointwise. Integrating in τ yields (7.G6).

Remark (how Corollary 7.GB is used to close P0.1). Corollary 7.GB converts a **single** pointwise gate opening into a **quantized** dissipation increment on a time-interval of fixed length. To turn this into the Gate Theorem (P0.1), one needs to show that such dissipation packets cannot occur

along the pinned RG orbit without violating the finite quantitative-scale dissipation budget (7.2). In the spine this is implemented by aligning the relevant gate radii R with the quantitative radii $\mathcal{R}_k$ on each window (so that $D_G(\cdot; 4R)$ dominates the per-step quantitative dissipation $\mathfrak{D}_k$). The required alignment is a bookkeeping statement rather than a new analytic estimate, but it must be written explicitly in §4.2.

6. Closure: finite dissipation forces rigidity (proof-spine polish)

This section turns the three pillars into the final contradiction.

6.1 What "finite dissipation" we actually need (and where it comes from)

Earlier drafts implicitly used a *monotone* full-space Gaussian budget to claim

$$\int_{-\infty}^{0} D_G[U](s)\, ds < \infty.$$

After the correction in §2.5, the full-space identity (6.3) contains the flux term $\mathcal{F}_G$, so monotonicity is not automatic and one should **not** bank the proof on it.

The rigidity step (M1: "infinitely many breaths force infinite dissipation") only needs **finite dissipation at the quantitative scale**. For the pinned orbit $\{U_k\}$, define the per-step quantitative dissipation

$$\mathfrak{D}_k \;:=\; \int_{-1}^{0} D_{4\mathcal{R}_k}[U_k](s)\, ds, \qquad \mathcal{R}_k := \mathcal{R}_{\theta,\kappa}[U_k(0)]. \tag{7.1}$$

What we will prove (using the now-closed M2 mechanism of §4.2.5–§4.2.6.5) is the summability

$$\sum_{k=1}^{\infty} \mathfrak{D}_k \;<\; \infty, \tag{7.2}$$

which is the exact "finite total dissipation" input needed to contradict an infinite sequence of breaths.

Proposition 7.1 (Finite quantitative-scale dissipation; cite-level). Let (U_k, P_k) be the pinned endgame windows with quantitative radii $\mathcal{R}_k$ and quantitative dissipation $\mathfrak{D}_k := \int_{-1}^{0} D_{4\mathcal{R}_k}[U_k](s)\, ds$. Then there exists a constant $c_* = c_*(\eta) > 0$ in the admissible constant class (Appendix NFS.1.9) such that for all $k \geq 1$,

$$E_{\mathcal{R}_k}[U_k](0) \;+\; c_*\, \mathfrak{D}_k \;\leq\; E_{\mathcal{R}_{k-1}}[U_{k-1}](0). \tag{7.3}$$

In particular, $\sum_{k=1}^{\infty} \mathfrak{D}_k < \infty$.

Proof (cite-level; no hidden hypotheses). Fix $k \geq 1$ and apply the fixed-cutoff identity (Lemma 7.10) on the unit window $[-1, 0]$ with radius $R = \mathcal{R}_k$:

$$E_{\mathcal{R}_k}[U_k](0) - E_{\mathcal{R}_k}[U_k](-1) + \int_{-1}^{0} D_{\mathcal{R}_k}[U_k](s)\, ds = \mathrm{Err}_{\mathcal{R}_k}[U_k; -1, 0].$$

By the signed trade (Option B, Lemma 7.1F + Lemma 7.1H together with Lemmas 7.1H0–7.1H1), there exist $\eta \in (0,1)$ and $C < \infty$ such that

$$\mathrm{Err}_{\mathcal{R}_k}[U_k; -1, 0] \le \eta \int_{-1}^{0} D_{4\mathcal{R}_k}[U_k](s)\, ds \;+\; C \Big(\sup_{s\in[-1,0]} \mathrm{Tail}_\sharp^{(k)}(s; \mathcal{R}_k/4) \Big)^2.$$

Since $D_{\mathcal{R}_k} \le D_{4\mathcal{R}_k}$ and $\mathfrak{D}_k = \int_{-1}^{0} D_{4\mathcal{R}_k}[U_k]$, we obtain (Proposition 7.1)

$$E_{\mathcal{R}_k}[U_k](0) + (1-\eta)\,\mathfrak{D}_k \le E_{\mathcal{R}_k}[U_k](-1) + C \Big(\sup_{s\in[-1,0]} \mathrm{Tail}_\sharp^{(k)}(s; \mathcal{R}_k/4) \Big)^2.$$

By the shift identity $U_k(\cdot, -1) = U_{k-1}(\cdot, 0)$, the left term on the right-hand side is $E_{\mathcal{R}_k}[U_{k-1}](0)$. The radii mismatch is handled in two deterministic steps.

1. **Log-Lipschitz dependence of E_R on the radius.** For any fixed profile V and any $R, R' \ge 1$,

$$|E_{R'}[V](0) - E_R[V](0)| \;\le\; C_{\mathrm{ER}}\, E_\infty[V](0)\, |\log(R'/R)|. \tag{7.3a}$$

where C_{ER} is an **absolute cutoff constant** depending only on the fixed cutoff template.

Proof sketch. Differentiate $E_{e^\rho}[V](0) = \int \frac{1}{2}|V|^2 \chi_{e^\rho}^2 G$ in ρ. The derivative is supported where $\nabla\chi \neq 0$ and satisfies $|\partial_\rho(\chi_{e^\rho}^2)| \lesssim 1$, hence (Proposition 7.1) $|\partial_\rho E_{e^\rho}[V](0)| \lesssim \int \frac{1}{2}|V|^2 G \le E_\infty[V](0)$. Integrate from $\rho = \log R$ to $\log R'$.

2. **Drift-control for the quantitative radius.** Along the pinned RG orbit, Lemma 7.9 combined with the stability estimate $\|\widetilde{F}_{-1} - \widetilde{F}_0\|_{L^\infty} \lesssim \mathfrak{D}_k$ yields a quantitative drift bound

$$|\log \mathcal{R}_k - \log \mathcal{R}_{k-1}| \;\le\; C_{\mathrm{drift}}\, \mathfrak{D}_k, \qquad C_{\mathrm{drift}} := \frac{C_{\mathrm{stab}}}{\kappa} \log R_+, \tag{7.3b}$$

where R_+ is the confinement radius from (7.29).

Combining (7.3a)–(7.3b) with $E_\infty[U_{k-1}](0) \le E_*$ (pin-time normalization), we obtain the comparison (Proposition 7.1)

$$E_{\mathcal{R}_k}[U_{k-1}](0) \;\le\; E_{\mathcal{R}_{k-1}}[U_{k-1}](0) \;+\; C_{\mathrm{mis}}\, \mathfrak{D}_k, \qquad C_{\mathrm{mis}} := C_{\mathrm{ER}}\, E_*\, C_{\mathrm{drift}}. \tag{7.3c}$$

Finally, we fix the band parameter κ (and then $\eta \in (0,1)$) at the outset so that

$$C_{\mathrm{mis}} \;\le\; \tfrac{1}{2}(1-\eta), \tag{7.3d}$$

which is possible since $C_{\mathrm{mis}} \propto \kappa^{-1}$ and κ is a free design parameter in the quantitative-radius definition (5.2). With this choice, the mismatch term is absorbed into $(1-\eta)\mathfrak{D}_k$, yielding $c_* := \frac{1}{2}(1-\eta) > 0$ in (7.3) (a fixed parameter constant once η is chosen; see Appendix NFS.1.9).

Finally, the No-Theft tail control (Lemma 7.8C_{NT}) implies the tail remainder tends to zero on the pinned tail, so summing (7.3) over k telescopes and yields $\sum_k \mathfrak{D}_k < \infty$.

Interpretation. The functional $E_{\mathcal{R}_k}$ is a Lyapunov-type quantity *at the quantitative scale*: each RG step either stabilizes the profile or it pays a definite amount of dissipation $\mathfrak{D}_k$. This is the correct replacement for any claim about monotonicity of the full-space E_G.

6.1.1 Historical checkpoint (resolved): why an absolute-value cutoff trade is insufficient

The fixed-cutoff endpoint identity (Lemma 7.10) yields, for any $R \geq 1$,

$$E_R(0) - E_R(-1) \;+\; \int_{-1}^{0} D_R(\tau)\, d\tau \;=\; \mathrm{Err}_R[U; -1, 0]. \tag{7.1A}$$

A purely **absolute-value** estimate of the form

$$|\mathrm{Err}_R[U; -1, 0]| \;\leq\; C \int_{-1}^{0} D_G[U](\tau; 4R)\, d\tau \;+\; C\Big(\sup_{\tau \in [-1,0]} \mathrm{Tail}_\sharp(\tau; R/4)\Big)^2 \qquad \text{(7.20C revisited)}$$

is therefore *not* sufficient, by itself, to force a **one-sided Lyapunov drop** at the quantitative scale (Proposition 7.1).

What is needed is a **signed absorption** with coefficient strictly smaller than 1, namely

$$\mathrm{Err}_R[U; -1, 0] \;\leq\; \eta \int_{-1}^{0} D_G[U](\tau; 4R)\, d\tau \;+\; C\Big(\sup_{\tau \in [-1,0]} \mathrm{Tail}_\sharp(\tau; R/4)\Big)^2, \qquad \eta < 1, \tag{RB1}$$

which, combined with (7.1A) and $D_R \leq D_{4R}$, yields

$$E_R(0) \;+\; (1-\eta) \int_{-1}^{0} D_G[U](\tau; 4R)\, d\tau \;\leq\; E_R(-1) \;+\; C\Big(\sup_{\tau \in [-1,0]} \mathrm{Tail}_\sharp(\tau; R/4)\Big)^2. \tag{7.1B}$$

Resolution (Option B, gauge-free). In the present canonical build, the signed trade (RB1) is proved *profile-free* by the transport–dichotomy mechanism (Option B) in §4.1.5: Lemma 7.1F (transport lower bound under absorption failure) and Lemma 7.1H (transport ⇒ Gate opening ⇒ dissipation packet), together with No-Theft, yield (RB1) on the pinned tail at the quantitative radii. Consequently, Proposition 7.1 and the quantitative budget (7.3) are unconditional in the present spine.

6.1.2 Route A executed: interior–exterior algebra and the remaining analytic nucleus (RB1→VI.RB1)

This subsection carries out **Route A** from §4.1.1 at the level of *explicit identities* and isolates the **single remaining estimate** needed to upgrade Proposition 7.1 to a fully proved Annals-level statement.

The guiding principle is simple:

- The fixed-cutoff identity (Lemma 7.10) is already available for the RG window class (thanks to local L^4 and commutator killing).
- The error Err_R is a sum of (i) annulus-supported commutator pieces and (ii) a *bulk Gaussian transport flux* induced by ∇G.
- Annulus pieces are already controlled by Lemma 7.8 (with a No-Theft vanishing remainder).
- Therefore, **RB1 reduces to showing that the bulk Gaussian transport flux can be absorbed into dissipation with coefficient < 1 up to a tail remainder.** (Proposition 7.1)

We implement this reduction as follows.

A. Interior / exterior cutoff family and the exact recombination Fix $R \geq 1$ and define the interior cutoff

$$\psi_R^{\text{in}} := \chi_R^2\, G.$$

To make the exterior test compactly supported for the local energy identity argument, introduce a second large radius $L \gg R$ and define the truncated exterior cutoff

$$\psi_{R,L}^{\text{out}} := (1 - \chi_R^2)\, \chi_L^2\, G.$$

Then $\psi_R^{\text{in}} \geq 0$, $\psi_{R,L}^{\text{out}} \geq 0$, and

$$\psi_R^{\text{in}} + \psi_{R,L}^{\text{out}} = \chi_L^2\, G. \tag{7.A1}$$

Apply Lemma 7.10 (localized energy **identity**, not just inequality) to ψ_R^{in} and $\psi_{R,L}^{\text{out}}$, add the two identities, and use (7.A1). One obtains the exact recombined identity (Proposition 7.1)

$$E_G[U](0; L) - E_G[U](-1; L) + \int_{-1}^{0} D_G[U](\tau; L)\, d\tau = \text{Err}_{\chi_L^2 G}[U; -1, 0], \tag{7.A2}$$

where the left-hand side uses the cutoff $\chi_L^2 G$ and the right-hand side is the corresponding error functional (as in Lemma 6.1 but with $\psi = \chi_L^2 G$).

Letting $L \to \infty$, and using the same tightness/pressure-localization bounds as in §2.5, yields the **full-space** identity on $[-1, 0]$:

$$E_G[U](0) - E_G[U](-1) + \int_{-1}^{0} D_G[U](\tau)\, d\tau = \mathcal{F}_G[U; -1, 0], \tag{7.A3}$$

with $\mathcal{F}_G$ as in §2.5. (At this stage we do **not** assume any sign for $\mathcal{F}_G$.)

Subtracting the (interior) fixed-cutoff identity (7.1A) from (7.A3) gives:

$$\text{Err}_R[U; -1, 0] = \mathcal{F}_G[U; -1, 0] - \Big(E_G(0) - E_G(-1)\Big) + \Big(E_R(0) - E_R(-1)\Big) - \int_{-1}^{0} \Big(D_G(\tau) - D_R(\tau)\Big)\, d\tau. \tag{7.A4}$$

This identity is purely algebraic. Its value is that all terms except the full-space flux $\mathcal{F}_G$ are *manifestly localized outside* B_R (they involve $G - \psi_R$ or $D_G - D_R$), hence are candidates for control by No-Theft/tightness (Proposition 7.1).

B. Isolation of the only remaining estimate (VI.RB1) Combining (7.A4) with Lemma 7.8 (which controls the annulus-supported pieces of Err_R by dissipation on scale $4R$ plus the vanishing tail remainder) reduces RB1 to the following:

VI.RB1 (Localized Gaussian bulk-flux absorption). There exist $\eta \in (0, 1)$ and $R_\star \geq 1$ such that for all $R \geq R_\star$,

$$\mathcal{F}_G[U;-1,0] \;\le\; \eta\int_{-1}^{0} D_G[U](\tau;4R)\,d\tau \;+\; C\Big(\sup_{\tau\in[-1,0]}\mathrm{Tail}_\sharp(\tau;R/4)\Big)^2, \tag{VI.RB1}$$

with $C = C(M, m_0, \chi)$.

Consequence. If (VI.RB1) holds, then the signed trade (RB1) follows (after choosing a fixed $R_\star \ge R_{\mathrm{NT}}$ large enough, per Appendix **NFS.1.8**, so that the tail remainder is below the required tolerance), hence Proposition 7.1 (RB1) becomes **proved** for the pinned tail (and therefore for all but finitely many windows) (Proposition 7.1).

Why this is the correct nucleus. - All annulus-supported cutoff/pressure commutators are already controlled by Lemma 7.8. - The only term that can still inject a *non-absorbable positive contribution* is the **bulk Gaussian flux** $\mathcal{F}_G$. - Showing $\mathcal{F}_G$ is absorbable with coefficient < 1 is strictly weaker than proving it is nonpositive, and the allowed remainder is precisely the No-Theft tail modulus.

C. Verification checklist (updated): what is certified at this stage The purpose of the preceding discussion is to isolate the *one-sided* requirement (RB1) from the purely absolute-value cutoff-trade estimates. In the present spine, (RB1) is **proved** (not assumed) by Option B in §4.1.5, using a profile-free transport dichotomy (Lemma 7.1F) and the Gate/No-Theft packet mechanism (Lemma 7.1H). No appeal is made to any monotonic full-space Gaussian budget.

Accordingly:

1. All previously stated modules (No-Theft, Gate barrier, Anti-Flash, packet charging) remain unchanged.
2. The signed absorption (RB1) is established on the pinned tail at the quantitative radii, with explicit constants.
3. Proposition 7.1 (quantitative budget) and the downstream "finitely many bad indices" conclusions in §4.2 are unconditional.

The remaining items after §4.1.5 are *presentation-level* only (index alignment and parameter bookkeeping), and are addressed explicitly in §4.2.

6.1.5.1 Annulus transport functional Fix a radius $R \ge 1$ and define the annulus $A_{4R} := \{4R < |y| < 8R\}$. Write the pressure in the gauge $\widetilde{P} := P - (P)_{B_{16R}}$ (Mini-Lemma 7.6P). Define the *annulus transport* functional

$$\Phi(R,\tau) := \frac{1}{R}\int_{A_{4R}} \Big(|U(\tau)|^3 + |U(\tau)|\,|\widetilde{P}(\tau)|\Big)\,G\,dy. \tag{7.1B0}$$

By CE1–CE3 (hence smoothness on the window; Lemma 7.GA), the map $\tau \mapsto \Phi(R,\tau)$ admits a continuous (RB1) representative on $[-1,0]$.

6.1.5.2 Lemma T: failure of strict absorption forces annulus transport The next lemma is the precise "transport detector" statement. It is formulated so that it can be inserted at the point where VI.RB1c is invoked: it turns the **absence of a strict constant** into a **positive transport integral**.

Lemma 7.1F (Transport lower bound from the absorption defect). Fix $\eta \in (0,1)$. Assume CE1–CE4 on the window $\tau \in [-1,0]$ and Uniform No-Theft (P0.2). Fix $R \geq 1$ and define the **absorption defect**

$$\mathrm{Def}_\eta(R) := \left[\mathcal{F}_G[U;-1,0] \;-\; \eta \int_{-1}^{0} D_G[U](\tau;4R)\,d\tau \;-\; C_{\mathrm{NT}} \Big(\sup_{\tau\in[-1,0]} \mathrm{Tail}_\sharp(\tau;R/4) \Big)^2 \right]_+ , \tag{7.1B1a}$$

The corresponding **strict absorption inequality** (equivalent to $\mathrm{Def}_\eta(R) = 0$) is

$$\mathcal{F}_G[U;-1,0] \;\leq\; \eta \int_{-1}^{0} D_G[U](\tau;4R)\,d\tau \;+\; C_{\mathrm{NT}} \Big(\sup_{\tau\in[-1,0]} \mathrm{Tail}_\sharp(\tau;R/4) \Big)^2 . \tag{7.1B1}$$

where C_{NT} is the constant in the far-field control of §4.1C (Step 1). Then there exist constants $C_{\mathrm{B}} > 0$ and $R_\star \geq 1$ (depending only on the window constants from CE and on η) such that for every $R \geq R_\star$,

$$\int_{-1}^{0} \Phi(R,\tau)\,d\tau \;\geq\; C_{\mathrm{B}}^{-1}\,\mathrm{Def}_\eta(R). \tag{7.1B2}$$

In particular, if the strict absorption inequality (7.1B1) fails at radius R, then $\mathrm{Def}_\eta(R) > 0$ and hence the annulus transport integral is strictly positive (Lemma 7.1F).

Moreover, by continuity of $\Phi(R,\cdot)$ (Lemma 7.GA) there exists $\tau_R \in [-1,0]$ such that

$$\Phi(R,\tau_R) \;\geq\; \int_{-1}^{0} \Phi(R,\tau)\,d\tau \;\geq\; C_{\mathrm{B}}^{-1}\,\mathrm{Def}_\eta(R), \tag{7.1B3}$$

Proof (cite-level). Write the bulk Gaussian flux as a near/far decomposition with the cutoff χ_{4R}:

$$\mathcal{F}_G[U;-1,0] = I_{\mathrm{near}}(R) + I_{\mathrm{far}}(R), \qquad I_{\mathrm{near}}(R) := \int_{-1}^{0}\!\!\int_{\mathbb{R}^3} \Big(\tfrac{1}{2}|U|^2 + P \Big)\, U \cdot \chi_{4R}^2 \nabla G\,dy\,d\tau.$$

By §4.1C (Step 1) and Uniform No-Theft (P0.2),

$$|I_{\mathrm{far}}(R)| \;\leq\; C_{\mathrm{NT}} \Big(\sup_{\tau\in[-1,0]} \mathrm{Tail}_\sharp(\tau;R/4) \Big)^2 . \tag{7.1B4}$$

Hence, by the definition (7.1B1a), (Lemma 7.1F)

$$\mathrm{Def}_\eta(R) \;\leq\; \Big[I_{\mathrm{near}}(R) \;-\; \eta \int_{-1}^{0} D_G[U](\tau;4R)\,d\tau \Big]_+ . \tag{7.1B5}$$

Reduction 7.1B6 (Defect → annulus transport). It suffices to bound the positive part on the right-hand side of (7.1B5) by the annulus transport functional.

It remains to bound the positive part on the right-hand side by the annulus transport functional. We work on B_{16R} and use the pressure gauge $\widetilde{P} := P - (P)_{B_{16R}}$ (Mini-Lemma 7.6P). Split $P = \widetilde{P} + (P)_{B_{16R}}$. The constant part is reduced to an **annulus term** by a cutoff integration by parts: writing $\chi_{4R}^2 \nabla G = \nabla(\chi_{4R}^2 G) - G\nabla(\chi_{4R}^2)$ and using $\nabla \cdot U = 0$, one has $\int U \cdot \nabla(\chi_{4R}^2 G)\, dy = 0$, hence (Lemma 7.1F) $(P)_{B_{16R}} \int U \cdot \chi_{4R}^2 \nabla G\, dy$ is supported where $\nabla \chi_{4R} \neq 0$ and is therefore (Lemma 7.1F) absorbed into the transport functional Φ (see §4.1C–§4.1E). Thus (Lemma 7.1F)

$$I_{\text{near}}(R) = \int_{-1}^{0} \int \left(\tfrac{1}{2}|U|^2 + \widetilde{P} \right) U \cdot \chi_{4R}^2 \nabla G\, dy\, d\tau.$$

Now apply the cutoff-trade identity from §4.1C–§4.1E (in particular §4.1E: the OU-cancellation and integration by parts that move derivatives onto χ_{4R}). Concretely, one writes $\chi_{4R}^2 \nabla G = \nabla(\chi_{4R}^2 G) - G\nabla(\chi_{4R}^2)$ and integrates by parts to place ∇ on $\frac{1}{2}|U|^2 + \widetilde{P}$. The bulk terms are controlled by dissipation using Cauchy–Schwarz and Young with a parameter $\varepsilon = \varepsilon(\eta)$ chosen so that the dissipation coefficient is $\leq \eta$; the only non-absorbed terms are those where ∇ hits χ_{4R}, hence are supported in A_{4R} and carry an explicit (Lemma 7.1F) R^{-1} factor. The outcome is the quantitative estimate

$$I_{\text{near}}(R) \;\leq\; \eta \int_{-1}^{0} D_G[U](\tau; 4R)\, d\tau \;+\; C_{\text{B}} \int_{-1}^{0} \Phi(R, \tau)\, d\tau, \tag{7.1B6}$$

where C_{B} depends only on the window constants and on η (through the Young parameter).

Combining (7.1B5) with (7.1B6) yields (7.1B2). The pointwise bound (7.1B3) follows from the trivial estimate $\sup_{\tau \in [-1,0]} \Phi(R, \tau) \geq \int_{-1}^{0} \Phi(R, \tau)\, d\tau$ (since $|[-1, 0]| = 1$). #### 6.1.5.3 Lemma G: transport activates the Gate and forces a dissipation packet

The next lemma is the conversion "transport Gate opening dissipation packet", which is the core of the censorship friction mechanism.

Lemma 7.1H (Transport triggers a quantized payment: Gate packet or dissipation packet). Fix the gate level $\gamma_0 > 0$ from §4.G and let δ_{gate} be the packet size (7.GP). There exist constants $\theta_0 = \theta_0(M, \gamma_0) > 0$, $D_0 = D_0(M, \gamma_0) > 0$, and $R_{\text{NT}} \geq 1$ such that for every $R \geq R_{\text{NT}}$ and every $\tau \in [-1, 0]$, if

$$\Phi(R, \tau) \;\geq\; \theta_0, \tag{7.1H1}$$

then **at least one** of the following two alternatives holds:

(H.a) *(Gate packet).* The cubic shell gate opens at scale $4R$:

$$\mathcal{G}(4R, \tau) \;:=\; \frac{1}{4R} \int_{A_{4R}} |U(\tau)|^3\, G\, dy \;\geq\; 2\gamma_0. \tag{7.1H2}$$

Consequently, by Lemma 7.GA and Corollary 7.GB, there exists a persistence interval $J \subset [-1, 0]$ with (7.GP) $|J| \geq c_\star(M, \gamma_0)$ such that

$$\int_J D_G[U](s; 16R)\, ds \;\geq\; \delta_{\text{gate}}. \tag{7.1H3}$$

(H.b) *(Direct dissipation packet).* One has the pointwise dissipation lower bound

$$D_G[U](\tau;16R) \;\geq\; D_0. \tag{7.1H4}$$

In this case (still under the transport-trigger regime (7.1H1)) the radius R lies in a compact range depending only on (M,γ_0), and the endpoint $L^\infty_\tau L^3_y$ regularity (Lemma 7.GA) yields a **profile-free** modulus of continuity for the map $s \mapsto D_G[U](s;16R)$. Concretely, Lemma 7.1H1 below provides a constant $c_{\rm diss} = c_{\rm diss}(M,\gamma_0) \in (0,1)$ such that there exists an interval $J \subset [-1,0]$ with $|J| \geq c_{\rm diss}$ on which $D_G[U](s;16R) \geq \frac{1}{2}D_0$. Therefore (Lemma 7.1H)

$$\int_J D_G[U](s;16R)\,ds \;\geq\; \tfrac{1}{2}D_0\,c_{\rm diss} \;=:\; \delta_{\rm diss}(M,\gamma_0). \tag{7.1H5}$$

In particular, whenever Uniform No-Theft makes $\mathrm{Tail}_\sharp(4R,\tau)$ sufficiently small, the threshold event (7.1H1) forces a **uniform, quantized payment** at scale $16R$:

$$\int_J D_G[U](s;16R)\,ds \;\geq\; \min\{\delta_{\rm gate},\delta_{\rm diss}\}. \tag{7.1H6}$$

Proof (cite-level). Split the transport functional $\Phi = \Phi_{\rm cub} + \Phi_{\rm prs}$ with

$$\Phi_{\rm cub}(R,\tau) := \frac{1}{R}\int_{A_{4R}} |U(\tau)|^3\,G\,dy, \qquad \Phi_{\rm prs}(R,\tau) := \frac{1}{R}\int_{A_{4R}} |U(\tau)|\,|\widetilde{P}(\tau)|\,G\,dy,$$

where $\widetilde{P} = P - (P)_{B_{8R}}$ (Mini-Lemma 7.6P; gauge conventions in Appendix NFS.1.6).

Step 1 (if the cubic part is large, the Gate opens). Since $\Phi_{\rm cub}(R,\tau) = 4\,\mathcal{G}(4R,\tau)$, the implication

$$\Phi_{\rm cub}(R,\tau) \;\geq\; 8\gamma_0 \quad\Rightarrow\quad \mathcal{G}(4R,\tau) \geq 2\gamma_0$$

is immediate, giving alternative (H.a).

Step 2 (pressure part: localization bound). Apply the pressure localization estimate (7.G2) with radius $4R$ (so that A_{4R} is the shell A_{4R} in the Gate notation and $B_{2\cdot 4R} = B_{8R}$ matches the gauge defining $\widetilde{P}$). This yields, for each fixed τ,

$$\Phi_{\rm prs}(R,\tau) \;\leq\; C\,\frac{1}{R}\int_{A_R^{32R}} |U(\tau)|^3\,G\,dy \;+\; C\,D_G[U](\tau;16R)^{1/2}\,\mathrm{Tail}_\sharp(4R,\tau). \tag{7.1H7}$$

(Here $A_R^{32R} := \{R \leq |y| \leq 32R\}$ is the union of dyadic shells A_{2^jR}, $j = 0,\dots,4$.)

Step 3 (either the Gate opens somewhere on the dyadic block, or the dissipation is large). Suppose (H.a) fails, i.e. $\mathcal{G}(4R,\tau) < 2\gamma_0$, and also assume the negation of (H.b), i.e. $D_G[U](\tau;16R) < D_0$. Then $\Phi_{\rm cub}(R,\tau) < 8\gamma_0$. Moreover, the first term on the right-hand side of (7.1H7) can be bounded by the dyadic gates on $[R,32R]$:

$$\frac{1}{R}\int_{A_R^{32R}} |U|^3\,G \;=\; \sum_{j=0}^{4}\frac{1}{R}\int_{A_{2^jR}} |U|^3\,G \;=\; \sum_{j=0}^{4} 2^j\,\mathcal{G}(2^jR,\tau). \tag{7.1H8}$$

If in addition all dyadic gates satisfy $\mathcal{G}(2^jR,\tau) \le 2\gamma_0$ for $j = 0, \dots, 4$, then the first term in (7.1H7) is $\le 2\gamma_0 \sum_{j=0}^{4} 2^j = 62\gamma_0$. For the second term we use $D_G[U](\tau; 16R)^{1/2} < D_0^{1/2}$, hence (Lemma 7.1H)

$$\Phi_{\mathrm{prs}}(R,\tau) \;\le\; 62C\,\gamma_0 \;+\; C\,D_0^{1/2}\,\mathrm{Tail}_\sharp(4R,\tau). \tag{7.1H9}$$

By Uniform No-Theft (P0.2) we may choose R_{pin} so large that $\sup_{\tau\in[-1,0]} \mathrm{Tail}_\sharp(4R,\tau) \le 1$ for all $R \ge R_{\mathrm{NT}}$. Fix $D_0 := \gamma_0^2$ and then choose

$$\theta_0 \;:=\; 8\gamma_0 \;+\; 62C\,\gamma_0 \;+\; C\,\gamma_0 \;+\; 1. \tag{7.1H10}$$

Under these choices, the conjunction "no Gate at $4R$, no large dissipation at $16R$, and no dyadic Gate on $[R, 32R]$" implies $\Phi(R,\tau) < \theta_0$, contradicting (7.1H1). Therefore, whenever (Lemma 7.1H) (7.1H1) holds, either (H.b) holds, or at least one dyadic gate $\mathcal{G}(2^jR,\tau)$ exceeds $2\gamma_0$ for some $j \in \{0, \dots, 4\}$.

Step 4 (reduce dyadic opening to the canonical scale $4R$). Since our transport functional $\Phi(R,\tau)$ is defined on the specific shell A_{4R}, the downstream use in Corollary 7.1I only requires a **quantized payment** at a radius comparable to R. If the dyadic opening occurs at $\rho := 2^jR$ with $j \ne 2$, we relabel the radius as $R \leftarrow \rho/4$; this preserves the form of the statement and changes the constants only by admissible dyadic factor (class (A))s. (This dyadic reindexing is formalized in Lemma 7.1H0 below.) In particular, we may assume the opening occurs at $4R$, giving (H.a), and then (7.1H3) follows from Lemma 7.GA and Corollary 7.GB.

The implication (H.b) $\Rightarrow$ (7.1H5) is proved in Lemma 7.1H1 below (closing VI.H1).

Corollary 7.1I (Defect threshold activates the Gate packet). Let C_{B} be as in Lemma 7.1F and let θ_0 be as in Lemma 7.1H. If, for some $R \ge R_\star$,

$$\mathrm{Def}_\eta(R) \;\ge\; C_{\mathrm{B}}\,\theta_0, \tag{7.1I1}$$

then there exists $\tau_R \in [-1,0]$ such that $\Phi(R,\tau_R) \ge \theta_0$. In particular, Lemma 7.1H yields a **quantized payment** on some interval $J \subset [-1,0]$ in the form (7.1H6) (after the dyadic reindexing (Lemma 7.1H0) if the opening occurs on a dyadic neighbor).

Proof. By Lemma 7.1F and (7.1B2), $\int_{-1}^{0} \Phi(R,\tau)\,d\tau \ge C_{\mathrm{B}}^{-1}\mathrm{Def}_\eta(R) \ge \theta_0$. Since $|[-1,0]| = 1$, there exists τ_R with $\Phi(R,\tau_R) \ge \theta_0$; then Lemma 7.1H applies.

6.1.5.4 How Option B is used in the canonical spine Verification items introduced by the cite-level conversion of Lemma 7.1H.

- **VI.H0 (Dyadic reindexing harmlessness; CLOSED in Lemma 7.1H0).** When a Gate opening is produced on a dyadic neighbor $\rho = 2^jR$ with $j \in \{0, \dots, 4\}$, we reparametrize $R \leftarrow \rho/4$ so that the opening is recorded at the canonical scale $4R$. This is a bookkeeping step: all shells/cutoffs in §4.1.5 use fixed universal dilation factors, hence dyadic relabeling changes constants by at most a factor of $2^{O(1)}$ (Corollary 7.1I) and does not alter any logical dependency.

- **VI.H1 (Uniform persistence for the dissipation alternative (H.b); CLOSED in Lemma 7.1H1).** Lemma 7.1H includes a "direct dissipation" alternative (7.1H4). In v5.9.93 we close the only remaining quantitative issue: we prove a **profile-free persistence length** $c_{\text{diss}}(M, \gamma_0) > 0$ (Lemma 7.1H1), which implies a **uniform** dissipation packet $\delta_{\text{diss}} = \frac{1}{2} D_0 c_{\text{diss}}$ whenever (H.b) occurs under the transport trigger. *(Optional simplification.)* One may still bypass the dissipation alternative entirely by using only the Gate-packet alternative (H.a) and strengthening the threshold so that the pressure term cannot generate $\Phi \geq \theta_0$ without opening a dyadic Gate; this is a presentation choice, not a logical necessity.

6.1.5.4.1 Closing VI.H0: dyadic reindexing We now close VI.H0 by making explicit the harmless dyadic relabeling used in Lemma 7.1H, Step 4.

Lemma 7.1H0 (Dyadic reindexing is harmless; closes VI.H0). Fix a base radius $R > 0$ and an index $j \in \{0, 1, 2, 3, 4\}$. Set the dyadic neighbor $\rho := 2^j R$ and define the relabeled radius $R' := \rho/4 = 2^{j-2} R$. Then:

1. (**Canonicalization of the shell.**) The geometric shell $A_{4R'}$ coincides with the shell A_ρ. Consequently, any (Lemma 7.1H0) functional defined by integrating on A_{4R} (such as $\Phi(R, \tau)$) can be evaluated at the neighbor scale by the relabeling $R \leftarrow R'$, i.e. $\Phi(R', \tau)$ is precisely the transport functional evaluated on the shell A_ρ.

2. (**Uniformity of cutoff bookkeeping.**) All cutoffs and localization radii used in §4.1.5 are built from R by **fixed universal dilation factors** (e.g. $2R, 4R, 16R, 32R$). Replacing R by R' multiplies each such radius by the same dyadic factor $2^{j-2} \in [2^{-2}, 2^2]$. Therefore (Lemma 7.1H0):

- the support relations (e.g. $\operatorname{supp} \nabla \chi_{4R} \subset A_{2R}$) are preserved verbatim after relabeling, and
- the size bounds on derivatives (e.g. $|\nabla \chi_{4R}| \lesssim 1/R$) change only by a **bounded** factor $\lesssim 4$.

3. (**Harmlessness for thresholds and packets.**) Any inequality in §4.1.5 of the schematic form

$$\text{LHS}(R) \;\leq\; C\,\text{RHS}(R)$$

with C depending only on the fixed dilation pattern remains valid after the relabeling $R \leftarrow R'$, with the constant multiplied by at most a admissible dyadic factor (class (A)) $2^{O(1)}$. In particular, when Lemma 7.1H produces a Gate opening at a dyadic neighbor $\rho = 2^j R$, we may **record** it at the canonical scale $4R'$ and invoke the corresponding Gate packet estimate (Corollary 7.GB / Lemma 7.2G) for R', without changing any logical dependency.

Proof (bounded dyadic change of scale).

The identity $A_{4R'} = A_\rho$ is immediate from $4R' = \rho$. The derivative support and size bounds follow because our cutoff profile is fixed once and for all and is rescaled by the radius; thus $|\nabla \chi_{cR'}| \sim 1/(cR')$ differs from (Lemma 7.1H0) $|\nabla \chi_{cR}| \sim 1/(cR)$ by the factor $R/R' \in [1/4, 4]$. All other occurrences of R in §4.1.5 enter only through such fixed dilation radii and the associated Gaussian weights on those shells, so every bound changes by at most a admissible dyadic factor (class (A)). Since the downstream spine uses only **quantized payment at a radius comparable to the trigger radius**, this dyadic relabeling is a purely bookkeeping step.

6.1.5.4.2 Closing VI.H1: uniform persistence for the dissipation threshold We now close VI.H1 by proving a **profile-free persistence** estimate for the localized Gaussian dissipation

$s \mapsto D_G[U](s;16R)$ in the regime where Lemma 7.1H is invoked (i.e. under the transport trigger (7.1H1)).

Lemma 7.1H1 (Uniform persistence for localized dissipation; closes VI.H1). Assume CE1–CE4 on an RG window $\tau \in [-1,0]$, and let $\theta_0, D_0, R_{\text{pin}}$ be as in Lemma 7.1H. Then there exists a constant

$$c_{\text{diss}} = c_{\text{diss}}(M,\gamma_0) \in (0,1) \tag{7.1H11}$$

with the following property. For every $R \geq R_{\text{NT}}$ and every $\tau_0 \in [-1,0]$ such that $\Phi(R,\tau_0) \geq \theta_0$ and $D_G[U](\tau_0;16R) \geq D_0$, there exists an interval $J \subset [-1,0]$ with $|J| \geq c_{\text{diss}}$ on which

$$D_G[U](s;16R) \;\geq\; \tfrac{1}{2}D_0 \qquad \text{for all } s \in J. \tag{7.1H12}$$

Consequently, $\int_J D_G[U](s;16R)\,ds \geq \delta_{\text{diss}} := \frac{1}{2}D_0\, c_{\text{diss}}$ (Lemma 7.1H1).

Proof (profile-free modulus of continuity under the transport trigger).

Step 1 (the transport trigger confines the radius). By CE1 and the Calderón–Zygmund (NFS.EST.4) bound $\|P\|_{L^{3/2}} \lesssim \|U\|_{L^3}^2$ (CE4), we have for each τ

$$\frac{1}{R}\int_{A_{4R}} (|U(\tau)|^3 + |U(\tau)|\,|\widetilde{P}(\tau)|)\,G\,dy \;\leq\; \frac{C}{R}\sup_{A_{4R}} G\,\|U(\tau)\|_{L^3}^3 \;\leq\; \frac{CM^3}{R}\,e^{-4R^2},$$

where we used $\sup_{|y|\geq 4R} G(y) \leq (4\pi)^{-3/2}e^{-(4R)^2/4} = Ce^{-4R^2}$. Therefore the threshold event $\Phi(R,\tau_0) \geq \theta_0$ implies $R \leq R_{\max}$, where $R_{\max} = R_{\max}(M,\theta_0)$ (Lemma 7.1H1) is the smallest radius for which $(CM^3/R)e^{-4R^2} < \theta_0$ holds for all $R \geq R_{\max}$. Since θ_0 depends only on (M,γ_0) (Lemma 7.1H), we may regard $R_{\max}$ as a function of (M,γ_0).

Step 2 (window regularity yields an L^2 control of $\partial_\tau \nabla U$). By Lemma 7.GA (endpoint $L^\infty_\tau L^3_y$ regularity), U is smooth on $(-1,0]\times\mathbb{R}^3$. Moreover, the quantitative form of this regularity on the fixed cylinder

$$Q_{\max} := [-1,0]\times B_{64R_{\max}}$$

provides a bound

$$\int_{-1}^{0}\!\!\int_{B_{64R_{\max}}} |\partial_\tau \nabla U|^2\,dy\,d\tau \;\leq\; K(M,\gamma_0)^2. \tag{7.1H13}$$

(Quantitative interior bounds are provided by Lemma 7.GB, which cites Zajączkowski [Zaj13, Thm. 4.6] and [Zaj13, Lem. 2.6]; see Appendix NFS.EST.7.)

Step 3 (modulus of continuity for ∇U in weighted L^2). Let $w_R(y) := \chi_{16R}(y)^2\,G(y)$. Since $R \leq R_{\max}$, we have $\operatorname{supp} w_R \subset B_{32R_{\max}}$ and $0 \leq w_R \leq 1$. For any $s \in [-1,0]$,

$$\|\nabla U(s) - \nabla U(\tau_0)\|_{L^2(w_R)} \;\leq\; \int_{\min\{s,\tau_0\}}^{\max\{s,\tau_0\}} \|\partial_\tau \nabla U(\sigma)\|_{L^2(B_{64R_{\max}})}\,d\sigma \;\leq\; |s-\tau_0|^{1/2}\,K(M,\gamma_0),$$

by Cauchy–Schwarz (NFS.EST.2) and (7.1H13).

Step 4 (choose a uniform persistence length). Write $D_G[U](\tau_0; 16R) = \|\nabla U(\tau_0)\|^2_{L^2(w_R)} \geq D_0$. If $|s - \tau_0| \leq c_{\text{diss}}$ and

$$c_{\text{diss}}^{1/2} K(M, \gamma_0) \;\leq\; \left(1 - 2^{-1/2}\right) \sqrt{D_0}, \tag{7.1H14}$$

then by the reverse triangle inequality,

$$\|\nabla U(s)\|_{L^2(w_R)} \;\geq\; \|\nabla U(\tau_0)\|_{L^2(w_R)} - \|\nabla U(s) - \nabla U(\tau_0)\|_{L^2(w_R)} \;\geq\; \sqrt{D_0} - \left(1 - 2^{-1/2}\right)\sqrt{D_0} \;=\; 2^{-1/2}\sqrt{D_0},$$

hence $D_G[U](s; 16R) \geq \frac{1}{2} D_0$. We therefore set (Lemma 7.1H1)

$$c_{\text{diss}} := \min\left\{1,\; \frac{\left(1 - 2^{-1/2}\right)^2 D_0}{K(M, \gamma_0)^2}\right\},$$

which depends only on (M, γ_0).

Step 5 (select an interval $J \subset [-1, 0]$ of length c_{diss}). If $\tau_0 \leq -c_{\text{diss}}/2$, take $J = [\tau_0, \tau_0 + c_{\text{diss}}]$. Otherwise take $J = [\tau_0 - c_{\text{diss}}, \tau_0]$. In both cases $J \subset [-1, 0]$, $|J| = c_{\text{diss}}$, and (7.1H12) holds by Step 4.

- **Benchmark (Option A):** §4.1.4 reduces strict absorption to VI.RB1c via OU-structure.
- **Intrinsic (Option B):** Lemma 7.1F + Lemma 7.1H show that *failure of strict absorption* forces a gate opening and therefore a **quantized** dissipation packet. On the RG orbit, repeated failures across indices k would then (Lemma 7.1H1) violate any finite quantitative budget once Proposition 7.1 is established.

In particular, once the quantitative dissipation summability $\sum_k \mathfrak{D}_k < \infty$ is proved (Proposition 7.1), Option B yields an explicit **index-level** asymptotic strict absorption statement on the pinned tail.

For each RG index $k \geq 1$, let $\mathcal{R}_k := \mathcal{R}_{\theta,\kappa}[U_k](0)$ and define the **Option B base radius**

$$R_k := \frac{\mathcal{R}_k}{4}. \tag{7.2B0}$$

This choice aligns the shells and radii used in §4.1.5 with the quantitative dissipation scale:

$$A_{4R_k} = A_{\mathcal{R}_k}, \qquad 16R_k = 4\mathcal{R}_k,$$

so that any quantized payment obtained from Lemma 7.1H at radius R_k is measured exactly by the per-step quantitative dissipation $\mathfrak{D}_k = \int_{-1}^{0} D_G[U_k](s; 4\mathcal{R}_k)\, ds$.

Write $\mathrm{Def}_\eta^{(k)}(R)$ for the absorption defect (7.1B1a) computed on the window U_k (with time variable $s \in [-1, 0]$). Then for any fixed threshold $\Theta > 0$, only finitely many indices can satisfy $\mathrm{Def}_\eta^{(k)}(R_k) \geq \Theta$, because each such event forces a **uniform dissipation packet** charged to $\mathfrak{D}_k$. Equivalently,

$$\mathrm{Def}_\eta^{(k)}(R_k) \to 0 \quad \text{as } k \to \infty,$$

i.e. the strict absorption inequality holds asymptotically along the RG orbit (and in particular, it holds eventually up to any prescribed tolerance). This implication is now written formally as Corollary 7.2J in §4.2.I.

6.2 The censorship friction mechanism, revisited

This section is the engine that turns **finite quantitative-scale dissipation** into convergence of modulation, hence into (Lemma 7.1H1) rigidity.

There are two steps:

1. **Local-to-quantitative:** prove that each genuine "breath" of the pinned profiles forces a quantitative lower bound on the per-step dissipation $\mathfrak{D}_k$ (this is M1, already written in §1.4, but it must be made purely analytic and cite-level).
2. **Global-to-finite:** prove the quantitative-scale budget (7.3), hence $\sum_k \mathfrak{D}_k < \infty$ (Proposition 7.1). This is where the cutoff-trade mechanism (M2a) and the drift-control conversion (M2b) enter.

The rest of §6 is devoted to step 2, with the goal of making the bound (7.3) completely cite-level.

We keep the notation of §1.1. For a RG-window U_k we write

$$\mathcal{R}_k(s) := \mathcal{R}_{\theta,\kappa}[U_k](s), \qquad \mathcal{R}_k := \mathcal{R}_k(0), \qquad \mathcal{R}_k^- := \mathcal{R}_k(-1).$$

By construction of the RG orbit (unit τ-translates, §1.1), $\mathcal{R}_k^-$ coincides with $\mathcal{R}_{k-1}$ (same physical time-slice, same gauge).

6.2.G Gate packets are charged to the quantitative budget (closing P0.1) We now write the bookkeeping step promised after Corollary 7.GB: a gate opening at the **quantitative radius** $\mathcal{R}_k$ forces a **uniform, quantized** contribution to the per-step quantitative dissipation $\mathfrak{D}_k$. Once this is written, P0.1 becomes a direct corollary of the finite quantitative budget (7.2).

Fix the gate threshold $\gamma_0 > 0$ and let $c_\star = c_\star(M, \gamma_0) \in (0,1)$ be the persistence length from Lemma 7.GA. Define the *packet size*

$$\delta_{\text{gate}} := \frac{\gamma_0}{C\,M}\, c_\star, \tag{7.GP}$$

where C is the constant in the cubic shell estimate (7.G1), hence also in (7.G3) (Lemma 7.1H1).

Lemma 7.2G (quantitative-scale gate opening ⇒ quantitative dissipation packet). Let $k \geq 1$. Assume that on the RG window $s \in [-1, 0]$ there exists $s_\star \in [-1, 0]$ such that

$$\mathcal{G}(\mathcal{R}_k, s_\star) \;\geq\; 2\gamma_0. \tag{7.2G1}$$

Then the per-step quantitative dissipation obeys

$$\mathfrak{D}_k \;=\; \int_{-1}^{0} D_{4\mathcal{R}_k}[U_k](s)\, ds \;\geq\; \delta_{\text{gate}}. \tag{7.2G2}$$

Proof. Apply Corollary 7.GB on the same window U_k with radius $R = \mathcal{R}_k$ and time $s_\star$. Then

$$\int_{s_\star}^{\min\{s_\star + c_\star, 0\}} D_G[U_k](s; 4\mathcal{R}_k)\, ds \;\geq\; \frac{\gamma_0}{C\,M}\, c_\star \;=\; \delta_{\text{gate}}.$$

Since $D_{4\mathcal{R}_k}[U_k](s)$ is exactly the localized Gaussian dissipation at radius $4\mathcal{R}_k$ (see the convention (7.R5) with $d\mu = G\,dy$), the left-hand side is bounded above by $\int_{-1}^{0} D_{4\mathcal{R}_k}[U_k](s)\,ds = \mathfrak{D}_k$, proving (7.2G2).

Corollary 7.2H (Only finitely many quantitative-scale gate openings). Assume the finite quantitative-scale dissipation budget (7.2): $\sum_{k\geq 1} \mathfrak{D}_k < \infty$. Then the set of indices

$$\mathcal{B} := \Big\{ k \geq 1 : \sup_{s\in[-1,0]} \mathcal{G}(\mathcal{R}_k, s) \geq 2\gamma_0 \Big\} \tag{7.2H1}$$

is finite, with

$$|\mathcal{B}| \;\leq\; \delta_{\text{gate}}^{-1} \sum_{k\geq 1} \mathfrak{D}_k. \tag{7.2H2}$$

Proof. For each $k \in \mathcal{B}$, Lemma 7.2G gives $\mathfrak{D}_k \geq \delta_{\text{gate}}$. Summing over $k \in \mathcal{B}$ yields $\delta_{\text{gate}}|\mathcal{B}| \leq \sum_{k\in\mathcal{B}} \mathfrak{D}_k \leq \sum_{k\geq 1} \mathfrak{D}_k < \infty$.

Theorem 7.GT (Gate closure near $\tau = 0$, in the pinned endgame). Assume (7.2). Then there exists $k_{\text{gate}} \in \mathbb{N}$ such that for all $k \geq k_{\text{gate}}$ and all $s \in [-1, 0]$,

$$\mathcal{G}(\mathcal{R}_k, s) \;<\; 2\gamma_0. \tag{7.GT}$$

In particular, in the pinned rigidity argument (which takes $k \to \infty$ to approach the putative singular time), the quantitative-scale gate can be treated as **uniformly closed**.

Proof. Take $k_{\text{gate}} := 1 + \max \mathcal{B}$, with the convention $k_{\text{gate}} = 1$ if $\mathcal{B} = \emptyset$. Then (7.GT) is exactly the negation of membership in $\mathcal{B}$.

Remark (what this closes, and what it does not). - The conclusion (7.GT) is the **exact** gate input needed downstream: every quantitative window in the pinned tail has the gate uniformly below the threshold, so the Gate barrier never needs to be invoked again in the endgame. - The stronger formulation "$\sup_{\tau\leq 0, R\geq 1} \mathcal{G}(R,\tau) \leq \gamma_0$" is strictly stronger than what the rigidity step consumes; if desired, it can be recovered by an additional (radius, time) covering argument that reduces arbitrary R to finitely many dyadic neighborhoods of the quantitative radii on the relevant tail interval.

6.2.I Option B defects are charged to the quantitative indices (explicit indexing) The Gate-charging Lemma 7.2G controls **quantitative-scale gate openings**. Option B (§4.1.5) produces such openings from a large absorption defect, but to avoid any ambiguity for an reader we now write the **exact index-level alignment** between the Option B radii and the per-step quantitative dissipation $\mathfrak{D}_k$.

Recall $\mathcal{R}_k := \mathcal{R}_{\theta,\kappa}[U_k](0)$ and set $R_k := \mathcal{R}_k/4$ as in (7.2B0). Define the **Option B packet size**

$$\delta_{\text{B}} := \min\{\delta_{\text{gate}},\, \delta_{\text{diss}}\}, \tag{7.2B1}$$

where δ_{gate} is the Gate packet size (7.GP) and δ_{diss} is the dissipation packet (7.1H5). Let

$$\Theta_{\mathrm{B}} := C_{\mathrm{B}}\,\theta_0, \tag{7.2B2}$$

where C_{B} is the transport constant from Lemma 7.1F and θ_0 is the trigger threshold from Lemma 7.1H.

Lemma 7.2I (Large Option B defect ⇒ quantitative dissipation packet). \n Fix $\eta \in (0,1)$. Assume CE1–CE4 and Uniform No-Theft on each RG window. Let R_{pin} be the radius from Lemma 7.1H. For each index $k \geq 1$ with $\mathcal{R}_k \geq 4R_{\mathrm{NT}}$ define $R_k := \mathcal{R}_k/4$ and let $\mathrm{Def}_\eta^{(k)}(R_k)$ be the absorption defect (7.1B1a) computed for the window U_k. If

$$\mathrm{Def}_\eta^{(k)}(R_k) \;\geq\; \Theta_{\mathrm{B}}, \tag{7.2I1}$$

then the per-step quantitative dissipation satisfies

$$\mathfrak{D}_k \;=\; \int_{-1}^{0} D_G[U_k](s;4\mathcal{R}_k)\,ds \;\geq\; \delta_{\mathrm{B}}. \tag{7.2I2}$$

Proof. Apply Corollary 7.1I to the RG-window U_k at the radius $R = R_k$. The assumption (7.2I1) gives $\Phi(R_k, \tau_{R_k}) \geq \theta_0$ for some $\tau_{R_k} \in [-1,0]$, hence Lemma 7.1H yields an interval $J \subset [-1,0]$ such that

$$\int_J D_G[U_k](s;16R_k)\,ds \;\geq\; \delta_{\mathrm{B}}.$$

Since $16R_k = 4\mathcal{R}_k$, the left-hand side is exactly $\int_J D_G[U_k](s;4\mathcal{R}_k)\,ds$, which is bounded above by $\int_{-1}^{0} D_G[U_k](s;4\mathcal{R}_k)\,ds = \mathfrak{D}_k$. This proves (7.2I2).

Corollary 7.2J (Only finitely many large Option B defects). \n Assume the finite quantitative dissipation budget (7.2): $\sum_{k\geq 1}\mathfrak{D}_k < \infty$. Then the set of indices

$$\mathcal{B}_{\mathrm{B}} := \Big\{k \geq 1 :\; \mathcal{R}_k \geq 4R_{\mathrm{NT}} \text{ and } \mathrm{Def}_\eta^{(k)}(R_k) \geq \Theta_{\mathrm{B}}\Big\} \tag{7.2J1}$$

is finite, with the quantitative bound

$$|\mathcal{B}_{\mathrm{B}}| \;\leq\; \delta_{\mathrm{B}}^{-1}\sum_{k\geq 1}\mathfrak{D}_k. \tag{7.2J2}$$

In particular,

$$\mathrm{Def}_\eta^{(k)}(R_k) \to 0 \quad \text{along any tail on which } \mathcal{R}_k \geq 4R_{\mathrm{NT}},$$

hence the strict absorption inequality holds asymptotically along the RG orbit at the aligned radii $R_k = \mathcal{R}_k/4$ (Corollary 7.2J).

Proof. For each $k \in \mathcal{B}_{\mathrm{B}}$, Lemma 7.2I gives $\mathfrak{D}_k \geq \delta_{\mathrm{B}}$. Summing over $k \in \mathcal{B}_{\mathrm{B}}$ yields $\delta_{\mathrm{B}}|\mathcal{B}_{\mathrm{B}}| \leq \sum_{k\in\mathcal{B}_{\mathrm{B}}}\mathfrak{D}_k \leq \sum_{k\geq 1}\mathfrak{D}_k < \infty$, proving (7.2J2).

Remark (why the restriction $\mathcal{R}_k \geq 4R_{\mathrm{NT}}$ is harmless). The constants θ_0 and R_{NT} in Lemma 7.1H are fixed once (M, γ_0) are fixed. Indices with $\mathcal{R}_k < 4R_{\mathrm{NT}}$ correspond to a uniformly bounded quantitative radius, i.e. a compact-core regime. Such indices can be handled separately (e.g. by

the gauge-fixed OU-coercivity route, or by a finite-cover argument in the compact radius range), and they do not affect the summability contradiction mechanism which is triggered by **infinitely many** large-defect events. #### 6.2.1 The width map is continuous in X

We will pass to limits in the window sequence. Since the width is defined through quantiles, we record a clean continuity statement for the *band-averaged* functional (5.2).

Lemma 7.2 (Continuity of $U \mapsto \mathcal{R}_{\theta,\kappa}[U](s)$).

Fix $s \in [-1, 0]$ and parameters (θ, κ). Assume $U^{(n)}(\cdot, s) \to U(\cdot, s)$ in the topology X, and that there exist constants $M_3 < \infty$, $m_1 > 0$ such that for all n,

$$\|U^{(n)}(\cdot, s)\|_{L^3(\mathbb{R}^3)} \le M_3, \qquad E_\infty[U^{(n)}](s) \ge m_1. \tag{7.2a}$$

Then

$$\mathcal{R}_{\theta,\kappa}[U^{(n)}](s) \to \mathcal{R}_{\theta,\kappa}[U](s).$$

Proof. For each fixed $R \ge 1$, the weight $\chi_R^2 G \in L^{3/2}$, and $X \hookrightarrow L^3_{\mathrm{loc}}$; hence $E_R[U^{(n)}](s) \to E_R[U](s)$. Similarly $E_\infty[U^{(n)}](s) \to E_\infty[U](s)$ by monotone convergence (Corollary 7.2J) in R combined with (7.2a). Therefore the normalized profiles (Corollary 7.2J) $\widetilde{F}_s^{(n)}(R) := E_R[U^{(n)}](s)/E_\infty[U^{(n)}](s)$ converge pointwise in R to $\widetilde{F}_s(R)$.

By Lemma 5.1A and (7.2a), all relevant quantiles satisfy $1 \le Q_s^{(n)}(u) \le R_+$ for $u \in [\theta - \kappa, \theta + \kappa]$ with the same R_+ for all n. A monotone-inverse stability lemma (for distribution functions; see e.g. Billingsley [Bil99, Thm. 14.2]) implies $Q_s^{(n)}(u) \to Q_s(\alpha)$ for a.e. α in $(0, 1)$. Since $|\log Q_s^{(n)}(u)| \le \log R_+$ on the band, dominated convergence yields

$$\int_{\theta-\kappa}^{\theta+\kappa} \log Q_s^{(n)}(u)\, d\alpha \;\to\; \int_{\theta-\kappa}^{\theta+\kappa} \log Q_s(\alpha)\, d\alpha,$$

and the claim follows from the definition (5.2). □

Lemma 7.GB (Quantitative interior derivative bound from the endpoint criterion). Let (U, P) be the LRNS orbit provided by Theorem 2.1. Fix a dyadic radius $R \ge 1$ and the annulus $A_R = \{R/2 \le |y| \le 4R\}$. Then there exists an admissible constant $C_{\mathrm{IB}} = C_{\mathrm{IB}}(M)$ such that

$$\sup_{\tau \in [-1,0]} \|\partial_\tau U(\tau)\|_{L^3(A_R)} \;\le\; C_{\mathrm{IB}}. \tag{7.GB}$$

(In particular $\tau \mapsto U(\tau)$ has a uniform modulus of continuity in $L^3(A_R)$ depending only on M on the unit window.)

Proof (cite-level; no spectral gap). By the endpoint regularity criterion at the $L^\infty_\tau L^3_y$ level (Escauriaza–Seregin–Šverák [ESS03] in the suitable framework, as used in [GKP13]), the LRNS orbit is smooth on $[-2, 0] \times A_{2R}$ once $\sup_\tau \|U(\tau)\|_{L^3} \le M$ is fixed. Quantitative interior estimates for the nonstationary Stokes system (in anisotropic Sobolev–Slobodetski classes, with localization to interior subdomains) are proved, for example, in Zajączkowski $[Zaj13, Thm.4.6]$. Moreover, the derivative interpolation inequality $[Zaj13, Lem.2.6]$ gives explicit control of time derivatives from the anisotropic norm. Applying these estimates on a slightly larger cylinder $[-2, 0] \times A_{2R}$ to the localized equation for ζU (with ζ supported in A_{2R} and $\zeta \equiv 1$ on A_R), and treating the nonlinear